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RESEARCH

Assessment of heavy metal contamination and ecological risk in alluvial sediments along the Mbam and Noun Rivers, Cameroon (SW Africa)

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Abstract This study presents the first comprehensive assessment of heavy metal contamination, ecological risk, and potential human health implications in alluvial sediments from the Mbam and Noun Rivers in Central Cameroon. Eighty-two sediment samples were collected and analyzed by inductively coupled plasma–mass spectrometry (ICP–MS) to determine the concentrations of Cr, V, Ni, Cu, Zn, Pb, Sn, U, and Th. The results were compared with upper continental crust (UCC) background values and ecotoxicological benchmarks, including threshold effect level (TEL), probable effect level (PEL), effect range—low (ERL), and effect range—median (ERM). Several contamination and ecological indices are as follows: contamination factor (CF), enrichment factor (EF), geo-accumulation index (Igeo), pollution load index (PLI), and ecological risk indices (Er, RI) were used to evaluate pollution intensity and its environmental implications. In the Mbam River, metal concentrations generally decreased upstream, following the order V > Cr > Zn > Ni > Cu > Pb > Th > Sn > U, with moderate contamination ($CF = 1.2–3.5$) and PLI values between 1.3 and 1.5, indicating diffuse anthropogenic influence. In contrast, the Noun River displayed extremely high uranium and vanadium enrichment ($CF_{(U)} > 5$; $EF_{(U)} \approx 3.8$), resulting in RI values up to 240, which denote considerable to very high ecological risk. Human health indices revealed hazard index (HI) values below 1 for all sites, indicating no immediate non-carcinogenic risk, though

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children showed higher exposure (HI_child = 0.19–0.34) than adults. Total carcinogenic risk (TCR) values (10^{-6}–10^{-4}) remained within the acceptable range but were highest at downstream Mbam sites, reflecting cumulative exposure to Cr (VI), Ni, and U. These findings indicate that metal enrichment in both rivers stems from the weathering of uranium-bearing lithologies combined with localized anthropogenic activities, including artisanal mining and agricultural runoff. Continuous monitoring and integrated watershed management are recommended to mitigate ecological degradation and long-term health risks in the Mbam–Noun River system.

Keywords Alluvial sediments · Cameroon rivers · Ecological risk assessment · Heavy metal contamination · Pollution indices · Uranium and vanadium pollution

Introduction

Sediment and soil geochemistry are essential to infer the primary source of potential mineralization, sediment provenance, environmental conditions, and heavy metal pollution (Afahnwie et al., 2025a; NdemaMbongué et al. 2025; Yiika et al., 2025). Aquatic environments are increasingly threatened by heavy metal contamination, which ranks among the most persistent and toxic pollutants found in sediments (Ali et al., 2019; Ekoa Bessa, 2024; Tiabou et al., 2024a). Fluvial sediments, particularly in alluvial contexts, act both as sinks and as sources of these contaminants, which may be remobilized depending on changes in environmental conditions (Ekoa Bessa et al., 2024; Kachoueiyan et al., 2024a; Salomons & Förstner, 1984; Simou et al., 2024a). Alluvial sediments are ecologically and economically vital and are most vulnerable to both natural and anthropogenic sources of pollution (Kachoueiyan et al., 2024b; Tiabou et al., 2025a; Tokatlı et al., 2025). The assessment of heavy metal concentrations in sediments is therefore a crucial step in diagnosing the environmental quality of river systems and understanding the long-term dynamics of pollutant accumulation (Chougong et al., 2022; Tokatlı & Islam, 2022; Sopie et al., 2022; Edjengte doumo et al., 2023).

The Mbam and Noun Rivers are two major tributaries of the Sanaga River, the primary hydrographic system in Cameroon. These watercourses traverse regions characterized by notable geological and socio-economic diversity, ranging from Precambrian basement rocks to intensive agricultural zones and peri-urban areas, making them potentially vulnerable to inputs of metals from both geogenic and anthropogenic sources (Désirée et al., 2021a, 2021b; Ekoa Bessa et al., 2022). The Mbam originates from the Adamawa Plateau and flows southward through the Centre Region, while the Noun rises in the West Region and joins the Mbam further east. These rivers provide essential ecosystem services, including water supply, irrigation, artisanal fishing, and sediment transport (Enock Embom et al., 2024; Kpoumié et al., 2020).

Over the past decades, rapid population growth, land-use changes, and artisanal exploitation activities have increased the risks of metal mobilization and sediment contamination in many river basins across sub-Saharan Africa (Dlamini et al., 2020; Edjengte doumo et al., 2022; NdemaMbongué et al., 2024; Kamguem Fotso et al., 2025). Cameroon is no exception, especially in the central and western regions, which face high anthropogenic pressure. Therefore, it is crucial to conduct integrated assessments that combine geochemical analyses with ecological risk indicators to evaluate the pollution levels in fluvial sediments.

This study aims to (1) determine the concentration levels of selected heavy metals (Cr, V, Ni, Zn, Cu, Pb, Sn, U, and Th) in the alluvial sediments of the Mbam and Noun Rivers; (2) assess the degree of contamination using contamination indicators; (3) estimate potential ecological risks through the risk factor (Er) and risk index (RI), following Hakanson's (1980) approach; and (4) identify the potential sources of heavy metals through multivariate approaches. The results will enhance understanding of the extent of pollution, its potential sources, and the ecological threats associated with these strategically important fluvial ecosystems of Central Africa.

Study area and geological setting

The Mbam and Noun Rivers are located in the Centre and West regions of Cameroon, respectively, in the southwestern part of Africa (Fig. 1a, b). There are two main tributaries of the Sanaga River, the largest

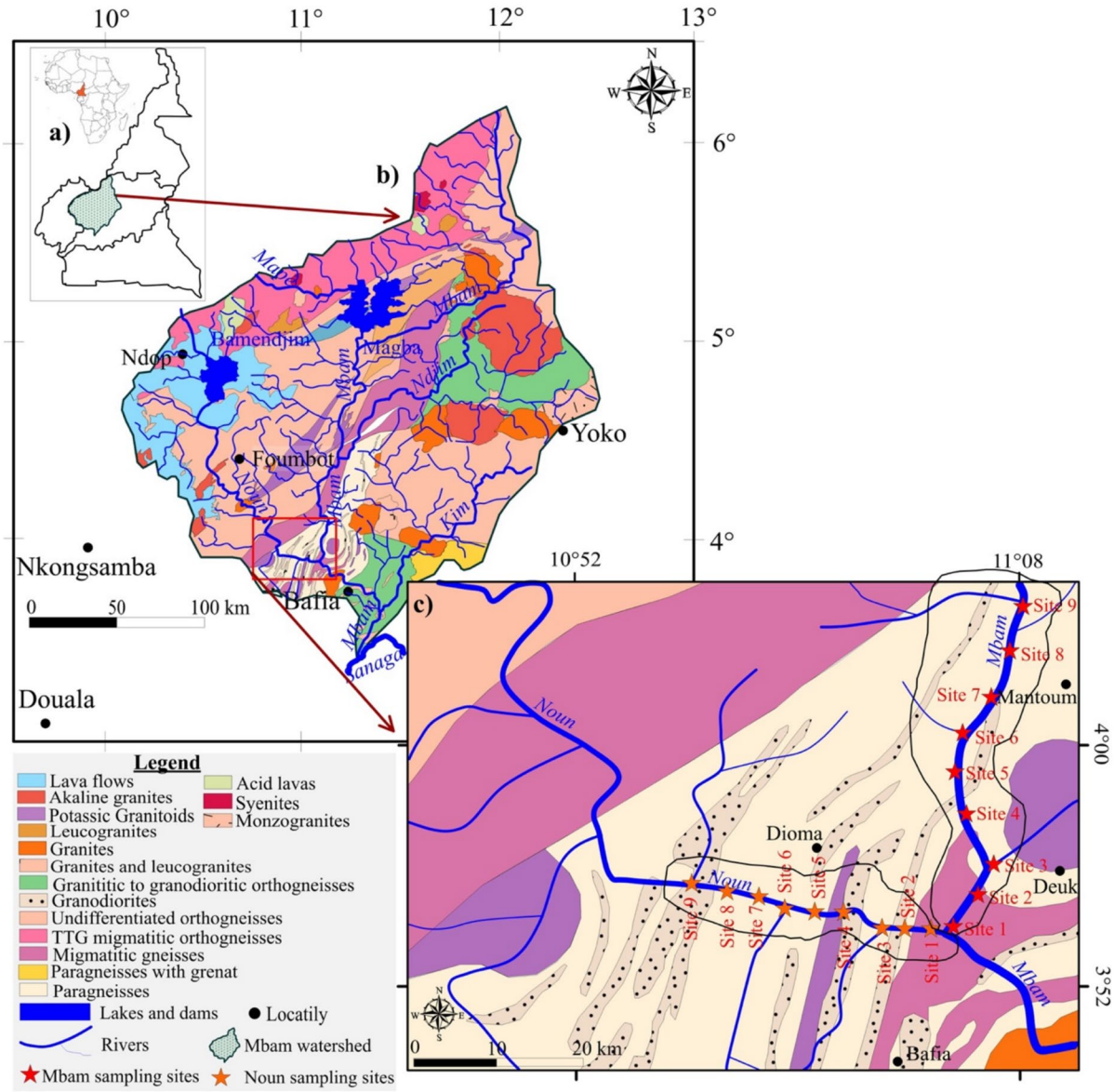


Fig. 1 Study area location. **a** Location of Cameroon in Africa, **b** simplified map of Mbam and Noun watershed and its geological formations (from Precasem, 2021), and **c** sampling sites location

hydrographic network in the country. The study area thus covers a large portion of the watersheds of these two rivers (Fig. 1b, c), which drain diverse landscapes ranging from high plateaus and deeply weathered terrains to intensively cultivated agricultural zones and peri-urban areas.

The Mbam River originates on the Adamawa Plateau at altitudes exceeding 1000 m and flows southward through the Centre Region before discharging into the Sanaga River near Ebebda. Its basin is dominated by Precambrian geological formations of the crystalline basement including gneisses, schists, amphibolites, and granitoids, which belong to the Pan-African Belt (Fig. 1b; Toteu et al., 2004; Precasem, 2021; Sigue et al., 2025a). Soils derived from these rocks are mainly ferralitic and highly weathered, making them significant contributors to the natural geochemical background of sediments transported by the river.

The Noun River, on the other hand, rises in the high plateaus of western Cameroon, near the town of Foumbot, and flows eastward until its confluence with the Mbam. Its basin is characterized by complex geology, consisting of Neoproterozoic basic complexes, volcanic formations, and more recent plutonic intrusions (Fig. 1b). In its upper reaches, the Noun drains high volcanic lands composed of basaltic and trachytic flows dating from the Miocene to Pleistocene, which belong to the Cameroon Volcanic Line (Fig. 1b). The lower reaches are dominated by Precambrian units similar to those found in the Mbam basin (Ngako et al., 2008; Precasem, 2021; Deffo et al., 2024; Ekoa Bessa et al., 2025; Valentino et al., 2025).

Both basins experience a humid tropical climate, characterized by a monsoonal to bimodal rainfall regime, an annual precipitation range of 1300 to 2000 mm, and a well-marked dry season from November to February. Heavy rainfall causes significant erosion, particularly in areas where vegetation cover is reduced due to agricultural practices or deforestation (Fonge et al., 2015). The Mbam basin is characterized by extensive and semicommercial agriculture. In contrast, the more densely populated Noun basin hosts notable agro-industrial activities and artisanal mineral exploitation, particularly around the towns of Foumbot, Bafia, and Magba (Mesmin et al., 2020; Precasem, 2021; Zébazé Togouet et al., 2023). These anthropogenic pressures may constitute additional sources of metals in sediments, supplementing the natural lithological geochemical background. The geological diversity, seasonal hydrodynamics, and increasing human disturbances make the Mbam and Noun Rivers ideal environments for studying heavy metal accumulation and their potential ecological impacts in tropical river systems of Africa.

Materials and methods

Sampling and preparation

The sampling campaign was conducted at a total of 18 sites, evenly distributed between the watersheds of the Mbam and Noun Rivers (Fig. 1c). At each site, between four and five composite surface sediment samples (0–5 cm) were collected using a clean plastic shovel to minimize any risk of contamination (Janpou et al., 2022). Sampling was carried out during the dry season to ensure optimal access to riverbanks and homogeneity in the fine alluvial deposits. The sediments were stored in clean, labelled polyethylene bags and transported to the laboratory. In the laboratory, the samples were air-dried at room temperature and then sieved to 2 mm to remove gravel and coarse organic debris. A representative fraction was subsequently ground using an agate mill until a fine, homogeneous powder (< 75 µm) was obtained for chemical analyses (Ekoa Bessa et al., 2018; Temgo Sopie et al., 2023).

Geochemical analyses of heavy metals

Chemical analyses were performed at the Geochemistry Laboratory of the Institute of Earth Sciences (ISTE), University of Lausanne (UNIL), Switzerland. Each sediment sample underwent total acid digestion ($HF–HNO_3–HClO_4$) in closed Teflon vessels resistant to high temperature and pressure. Total concentrations of metallic elements (Cr, V, Ni, Zn, Cu, Pb, Sn, U, Th) were determined by inductively coupled plasma–mass spectrometry (ICP–MS) using a Thermo Scientific iCAP-Q spectrometer (Ekoa Bessa et al., 2024; Kamguem Fotso et al., 2025). Each sample was analyzed in triplicate, and the reported concentrations represent the arithmetic mean of these three independent measurements. Quality assurance and quality control (QA/QC) procedures were strictly followed throughout the analytical process. Analytical blanks were systematically included after every 10 measurements to monitor potential contamination, and certified reference materials (SRM 2711a and SO-18) were analyzed under the same conditions to verify instrument accuracy. The analytical precision and accuracy were better than ±10%, and reproducibility exceeded 95% for all measured elements.

Instrument calibration was performed using multielement standard solutions, and the correlation coefficients (R^2) of calibration curves were consistently greater than 0.999. The method detection limits (MDLs), determined as three times the standard deviation of procedural blanks, were below 0.01 $mg \cdot kg^{-1}$ for all analyzed elements, ensuring high analytical sensitivity. All analytical operations were carried out under clean laboratory conditions to minimize contamination risk, and data quality was verified through internal consistency checks and cross-validation of replicate analyses.

Calculation of contamination indices

To assess the degree of heavy metal contamination in sediments, several geochemical indices were calculated based on the measured concentrations, using background values from the upper continental crust (UCC) as defined by Rudnick and Gao (2003). These indices, described below, allow for the distinction between enrichments of natural (geogenic) origin and those resulting from human (anthropogenic) activities.

Contamination factor (CF)

The contamination factor (CF) is one of the fundamental indicators used to quantify metal pollution in sediments (Yiika et al., 2022). It is defined as the ratio between the measured concentration of a given metal in a sample (C_n) and its estimated geochemical background value (Rudnick & Gao, 2003) for the upper continental crust (*Cref*). The mathematical expression of this factor is given by Eq. 1 (Hakanson, 1980):

$$CF = \frac{C_n}{C_{Ref}} \tag{1}$$

This index makes it possible to assess the extent to which an element is enriched relative to its expected natural background level. Its interpretation is based on threshold classes: a CF value below 1 indicates low contamination or no enrichment, a CF between 1 and 3 denotes moderate contamination, values between 3 and 6 correspond to considerable contamination, whereas a CF greater than or equal to 6 reflects very high contamination, possibly due to significant anthropogenic inputs NdemaMbongué et al., 2023; Tiabou et al., 2024b) (Table 1). The CF is therefore a simple and robust tool for a preliminary evaluation of the pollution level specific to each metal studied.

Geo-accumulation index (Igeo)

The geo-accumulation index (Igeo), initially proposed by Muller (1969), aims to refine the assessment of sediment pollution by incorporating a margin for natural variability through an empirical correction factor (1.5). This index is calculated using the following formula, given in Eq. 2:

$$Igeo = log_2\left(\frac{C_n}{1.5 \times C_{Ref}}\right) \tag{2}$$

where C_n is the observed concentration in the sample and *Cref* is the background concentration (Rudnick & Gao, 2003). The interpretation of Igeo is based on a semiquantitative seven-class scale: an Igeo ≤ 0 indicates the absence of pollution, values between 0 and 1 correspond to low pollution, 1 to 2 indicates moderate

Table 1 Classes of pollution indices (EF, I-geo, RI, CF, and PLI) and contamination levels used in this study

EF classes	Enrichment level	I-geo value, classes	Pollution level
$EF < 1$	No enrichment	I-geo ≤ 0, 0	Unpolluted
$EF = 1–3$	Minor enrichment	I-geo = 0–1, 1	Unpolluted to moderately polluted
$EF = 3–5$	Moderate enrichment	I-geo = 1–2, 2	Moderately polluted
$EF = 5–10$	Moderately severe enrichment	I-geo = 2–3, 3	Moderately to strongly polluted
$EF = 25–50$	Very severe enrichment	I-geo = 3–4, 4	Strongly polluted
$EF > 50$	Extremely severe enrichment	I-geo = 4–5, 5	Strongly to very strongly polluted
RI classes	Risk level	CF classes	Contamination level
$RI < 150$	Low ecological risk	$CF < 1$	Low contamination
$RI = 150–300$	Moderate ecological risk	$CF = 1–3$	Moderate contamination
$RI = 300–600$	Significant ecological risk	$CF = 3–6$	Considerable contamination
$RI > 600$	High ecological risk	$CF > 6$	High contamination
PLI, classes	Sediment quality	EF, The enrichment factor (Rudnick & Gao, 2003); Igeo, the geo-accumulation index (Muller, 1969); CF, the contamination factor (Hakanson, 1980; Tomlinson et al., 1980); PLI, the pollution load index (Tomlinson et al., 1980); RI: The Potential Ecological Risk index (Hakanson, 1980)	
$PLI < 1$	No pollution		
$PLI > 1$	Polluted		

pollution, 2 to 3 denotes moderately high pollution, 3 to 4 indicates high pollution, 4 to 5 denotes very high pollution, and Igeo > 5 reveals extremely high pollution (Suh et al., 2025) (Table 1). This index is particularly relevant in contexts where natural fluctuations in the geochemical background must be considered to distinguish anomalies of anthropogenic origin (Tiabou et al., 2025c).

Enrichment factor (EF)

The enrichment factor (EF) is used to detect abnormal enrichment by comparing the relative concentration of a metal to that of a lithogenic reference element in both the sample and the continental crust. The EF is calculated according to the formula given in Eq. 3 (Buat-Menard & Chesselet, 1979):

$$EF = \left(\frac{C_n/Th}{C_{Ref}/Th_{Ref}}\right) \quad (3)$$

where *Th* represents the concentration of thorium in the sample, used here as the normalizing element, and *Thref* is its background value. This factor helps identify potential anthropogenic inputs by distinguishing natural enrichments from those of human origin. The commonly adopted interpretation scale is as follows: an EF below 1.5 indicates a predominantly natural origin; values between 1.5 and 3 suggest minor enrichment of human origin, 3 to 5 corresponds to moderate enrichment, 5 to 10 indicates considerable enrichment; and an EF greater than 10 reflects very high enrichment, strongly suspected to be of anthropogenic origin (Simou et al., 2024b; Tiabou et al., 2024c) (Table 1). In this respect, EF is an essential tool for assessing the degree of disturbance in aquatic environments caused by human activities while taking into account the local geochemical background (Tiabou et al., 2025c; Sigue et al., 2025a).

The choice of thorium (Th) as the reference element for calculating the enrichment factor is based on its remarkably stable and conservative geochemical properties, as established in fundamental studies of crustal composition (Taylor & McLennan, 1985; Wedepohl, 1995). Thorium exhibits low mobility under normal oxic conditions, with studies demonstrating limited transport in natural waters at low temperatures (Langmuir & Herman, 1980). Furthermore, Th is minimally influenced by anthropogenic inputs, making it a reliable indicator for distinguishing natural from anthropogenic sources (Reimann & de Caritat, 2005), and serves as a good representative of the lithogenic (mineral-derived) fraction of sediments in geochemical assessments (Audry et al., 2004; Salminen et al., 2005). Moreover, Th is less sensitive to grain-size effects than other commonly used normalization elements such as iron (Fe) or aluminum (Al) (Covelli & Fontolan, 1997; Loring, 1991), which helps reduce biases related to fine clay content or sediment sorting processes that can affect normalization procedures (Rubio et al., 2000; Schropp et al., 1990).

By comparison, Fe is abundant and easy to analyze, but its concentration can be significantly altered by redox conditions and by anthropogenic inputs, particularly from industrial sources. Al is a good crustal marker, but its distribution is strongly correlated with clay content, which may introduce substantial granulometric bias. Titanium (Ti), although conservative and relatively immobile, may be locally enriched in heavy minerals such as ilmenite, compromising its geochemical representativity. Thorium, while rarely affected by anthropogenic sources, may display some natural variability depending on the local lithological context. Nevertheless, its generally conservative behavior makes it a reference element of choice in environmental geochemical studies involving varied sediment grain sizes, such as riverbank deposits. This position is supported by several reference studies in the scientific literature, including those by Loring and Rantala (1992), McLennan (2001), and Lima et al. (2005), which have demonstrated the relevance and robustness of using Th as a normalizing element in the assessment of anthropogenic enrichment in heavy metals.

Pollution Load Index (PLI)

The Pollution Load Index (PLI) is a synthetic tool used to assess the overall level of metal pollution in a given sample by taking into account all the analyzed heavy metals (Sigue et al., 2025b; Yiika et al., 2024). Unlike element-specific indices, the PLI aims to provide an integrated overview of site contamination by incorporating the contribution of multiple pollutants through a geometric approach. The calculation of the PLI is based on the contamination factors (CF) determined for each metal (see the section "Contamination factor (CF)", Eq. 1). It is mathematically expressed as

the geometric mean of the product of the CF values for all considered metals, as given in Eq. 4 (Tomlinson et al., 1980):

$$PLI = (CF_1 \times CF_2 \times \cdots \times CF_n)^{1/n} \quad (4)$$

where n represents the total number of metallic elements studied. This index has the advantage of summarizing, in a single value, the overall pollution load of a given site while accounting for the individual variations of each metal. The interpretation of the PLI is straightforward: a value equal to 1 indicates that the average metal concentrations are comparable to the natural geochemical background (i.e., no significant pollution), a $PLI < 1$ reflects an absence of contamination or concentrations below reference levels, whereas a $PLI > 1$ indicates significant site-level pollution (Table 1), potentially due to anthropogenic inputs. The main interest of the PLI lies in its ability to compare different sites or areas within a watershed, identifying the sectors most impacted by metal pollution. It is a widely used indicator in recent environmental studies for the monitoring of fluvial, harbor, or lacustrine sediments.

Ecological risk assessment

The assessment of ecological risk associated with the presence of heavy metals in sediments is based on the approach developed by Hakanson (1980), which is widely applied in contemporary environmental studies. This method combines information on metal concentrations, their contamination levels relative to a natural geochemical background, and their potential toxicity to aquatic organisms. It thus allows for quantifying the hazard posed by each element to the environment, as well as estimating the cumulative impact at the site scale. The fundamental index used in this method is the potential individual ecological risk factor (Er). This indicator combines the measured concentration of a metal with its relative toxicity through the calculation given in Eq. 5:

$$Er_i = Tr_i \times CF_i \quad (5)$$

where CFi is the contamination factor of metal i (ratio between the measured concentration and the background concentration, see the section "Calculation of contamination indices") and Tri is the toxicity factor assigned to the metal, reflecting its potential to harm aquatic organisms (Afahnwie et al., 2025b). These toxicity factors were empirically defined by Hakanson (1980) and have been adopted in numerous subsequent studies. For example, the generally accepted values are as follows: $Cr = 2$, $Ni = 5$, $Cu = 5$, $Zn = 1$, $Pb = 5$, $V = 1$, $Sn = 1$, $U = 30$, and $Th = 5$.

The interpretation of the Er index is as follows: an $Er < 40$ indicates low ecological risk, an Er between 40 and 80 suggests moderate risk, an Er between 80 and 160 corresponds to high risk, and an $Er > 160$ signals very high ecological risk for the local aquatic environment (Table 1).

To move beyond the individual element analysis, the Hakanson (1980) approach also proposes the calculation of the overall ecological risk index (RI), obtained by summing the Eri values for all metals analyzed in a given sample, according to Eq. 6:

$$RI = \sum Er_i \quad (6)$$

This index is used to characterize the overall level of potential ecological stress at a given site. The interpretation proposed by Hakanson distinguishes several risk classes: an $RI < 150$ indicates a low overall risk, an RI between 150 and 300 corresponds to moderate risk, an RI between 300 and 600 reflects considerable risk, and an $RI > 600$ signals a very high ecological risk (Table 1). The strength of this integrated approach lies in its twofold nature: it accounts for both the observed contamination levels and the specific hazards posed by each metal while also enabling the ranking of sites according to their level of ecological vulnerability. It is therefore particularly well-suited for risk assessment in tropical river systems where natural and anthropogenic sources of metal pollution coexist.

Human health risk assessment

To complement the ecological risk evaluation, the potential human health risks associated with heavy metal contamination in sediments were quantified using both non-carcinogenic and carcinogenic exposure models following the framework of the United States Environmental Protection Agency (USEPA, 1989, 2011). The exposure pathway considered was the incidental ingestion of contaminated sediments, representing the main route of exposure for local

populations living near riverbanks (Li et al., 2014; Wang et al., 2020). The non-carcinogenic risk was assessed through the computation of the chronic daily intake (CDI), the hazard quotient (HQ), and the hazard index (HI), using Eq. 7:

$$CDI = \frac{C \times IR \times EFr \times ED \times CFa}{BW \times AT} \quad (7)$$

where C (mg·kg^{-1}) is the concentration of the metal in sediments, IR (mg·day^{-1}) is the ingestion rate, EFr (days·year^{-1}) the exposure frequency, ED (years) the exposure duration, CFa (10^{-6} kg·mg^{-1}) the conversion factor, BW (kg) the body weight, and AT (ED × 365 days) the averaging time for non-carcinogenic effects (USEPA, 1989, 2011). Default residential parameters were adopted: children ($IR = 200$ mg·day^{-1}, $BW = 15$ kg, $EFr = 350$ days·year^{-1}, $ED = 6$ years) and adults ($IR = 100$ mg·day^{-1}, $BW = 70$ kg, $EFr = 350$ days·year^{-1}, $ED = 24$ years) (Li et al., 2014; Shao et al., 2021). The HQ for each metal was calculated as $HQ = \text{CDI/RfD}$, where RfD (mg·kg^{-1}·day^{-1}) values were taken from the USEPA Integrated Risk Information System (USEPA IRIS, 2023): Cr (III) = 1.5, $V = 0.009$, $Ni = 0.02$, $Zn = 0.3$, $Cu = 0.04$, $Sn = 0.3$, $U = 0.003$. Lead (Pb) and thorium (Th) were excluded due to the absence of reliable RfD values in IRIS. The HI was then computed as the sum of HQ values for all metals ($HI = \Sigma\text{HQ_i}$). A $HI < 1$ indicates no expected adverse health effects, whereas $HI > 1$ suggests potential non-carcinogenic risk (Muhammad et al., 2021; USEPA, 2011).

The carcinogenic risk (CR) was estimated for metals with known carcinogenic potential Cr (as Cr (VI)), Ni, and U using the equation $CR = \text{CDI} \times \text{SF}$, where SF ((mg·kg^{-1}·day^{-1}) is the cancer slope factor obtained from USEPA (IRIS, 2023) and other published databases (Wu et al., 2022; Yousaf et al., 2024): Cr (VI) = 0.5, $Ni = 0.84$, and $U = 1.5 \times 10^{-5}$. The total carcinogenic risk (TCR) was calculated as $TCR = \Sigma\text{CR_i}$, and lifetime exposure was considered with an averaging time of 70 years × 365 days (USEPA, 1989). TCR values between 10^{-6} and 10^{-4} were interpreted as acceptable according to USEPA standards (Li et al., 2014; USEPA, 2011). All computations were performed separately for children and adults using Python 3.11 and Microsoft Excel 16. This approach provides a quantitative estimation of both immediate- and long-term human health effects due to metal exposure through sediments, enabling comparison between sites and identification of the most hazardous elements within the Mbam and Noun River systems (Shao et al., 2021; Wang et al., 2020).

Statistical analyses

Geochemical data were subjected to advanced statistical processing to explore relationships between the different heavy metals and to better understand potential contamination sources. A descriptive analysis, including the calculation of means, standard deviations, and coefficients of variation, was first carried out to characterize the spatial distribution of concentrations. Subsequently, a principal component analysis (PCA) was applied to reduce the dimensionality of the dataset and group metals exhibiting similar behaviors, thereby facilitating the interpretation of geochemical processes and the respective influences of natural and anthropogenic inputs. Pearson correlation coefficients were calculated to quantify linear relationships between the concentrations of different elements (Afahnwie et al., 2025b; Ekoa Bessa, 2024; Kouankap et al., 2024). In addition, a hierarchical cluster analysis was performed, resulting in the generation of dendrograms. This method allowed for grouping samples and/or metals according to their geochemical similarity, providing an additional tool for identifying sources and processes of contamination. All statistical analyses were conducted using R software (version 4.x) and the Python environment (pandas, NumPy, Matplotlib, and scikit-learn libraries), with a significance threshold set at $p < 0.05$.

Results

Heavy metal concentrations

The concentrations of heavy metals (Cr, V, Ni, Zn, Cu, Pb, Sn, U, and Th) in sediments from the Mbam and Noun Rivers show significant spatial variability along the upstream–downstream gradient (Table 2; Fig. 2). In the Mbam River, Cr, Ni, Zn, Cu, and Pb concentrations are generally higher at sites near the river mouth (Sites 1 to 3), with values reaching 143 mg/kg for Cr, 42 mg/kg for Ni, 86 mg/kg for Zn, and 35 mg/kg for Cu (Table 2, Fig. 2). These concentrations tend to gradually decrease upstream, with the most distant sites (Sites 8 and 9) showing lower

levels (e.g., *Cr*≈88–90 mg/kg, *Zn*≈55–65 mg/kg). This trend suggests greater heavy metal accumulation downstream, resulting from a combination of natural geogenic contributions, mainly the weathering of gneissic, amphibolitic, and uraniferous granitoid formations that dominate the local basement, and additional anthropogenic inputs from agricultural runoff and fluvial convergence processes.

In contrast, some metals, notably vanadium (V) and thorium (Th), display a more homogeneous distribution along the Mbam, with average concentrations ranging from 95 to 198 mg/kg for V and 16 to 22 mg/kg for Th (Table 2; Fig. 2), without a clear spatial gradient. Uranium (U) concentrations remain low in the Mbam (≈4–5 mg/kg), with no significant variation between upstream and downstream locations.

In the Noun River, results show a different profile. Uranium (U) and vanadium (V) concentrations are remarkably high along the entire river, reaching 15–19 mg/kg and 175–185 mg/kg, respectively (Table 2; Fig. 2), without a marked upstream decrease. Other metals, notably Cr, Ni, Zn, Cu, and Pb, occur at much lower levels than in the Mbam (e.g., *Cr*≈4–8 mg/kg, *Ni*≈4–5 mg/kg), which may reflect a lower anthropogenic influence or a different geochemical background. Thorium (Th) remains relatively constant (≈20 mg/kg) along the Noun.

A comparison between the two rivers highlights marked differences: the Mbam is characterized by generally higher concentrations of most heavy metals, especially in its downstream sectors, while the Noun stands out for its very high levels of vanadium and uranium, evenly distributed along the river before its confluence with the Mbam. This spatial distribution suggests predominantly geogenic sources for V and U in the Noun River, derived from the weathering of uraniferous granites, migmatitic gneisses, and amphibolites from the Mbam-Noun catchment geology (Precasem, 2021; Sigue et al., 2025a). Vanadium and uranium are released from mafic silicates and accessory minerals such as biotite, amphibole, and uraninite during

Table 2 Average concentrations of trace metals (mg·kg^{-1}) in individual surface sediment samples from the Mbam and Noun rivers, compared with geochemical and ecotoxicological reference values

Rivers	Sites (samples)	Cr	V	Ni	Zn	Cu	Pb	Sn	U	Th
MBAM	Site 1 ($n=4$)	109.20	124.80	42.60	71.20	35.40	30.90	8.60	5.50	21.20
	Site 2 ($n=5$)	89.30	98.50	32.60	60.30	26.10	27.60	9.10	4.50	21.79
	Site 3 ($n=4$)	143.00	198.30	42.00	86.60	32.00	23.80	4.70	2.80	16.20
	Site 4 ($n=5$)	126.10	161.50	42.30	78.90	33.70	27.30	6.60	4.10	18.70
	Site 5 ($n=4$)	116.90	145.70	39.80	74.20	31.80	27.40	7.20	4.20	19.40
	Site 6 ($n=4$)	89.50	98.50	32.60	60.30	26.10	27.60	9.00	4.55	21.80
	Site 7 ($n=5$)	107.80	130.00	37.45	69.60	29.90	27.45	7.80	4.32	20.25
	Site 8 ($n=5$)	87.90	102.40	34.40	65.00	26.80	27.10	8.30	4.60	22.20
	Site 9 ($n=5$)	90.20	94.60	30.90	55.70	25.40	28.10	9.70	4.50	21.40
Noun	Site 1 ($n=5$)	3.90	178.30	4.80	28.10	5.40	30.20	7.10	15.40	20.60
	Site 2 ($n=5$)	7.90	180.60	4.60	26.50	5.30	28.50	6.90	19.30	18.70
	Site 3 ($n=4$)	5.90	179.40	4.69	27.30	5.30	29.30	7.40	17.30	19.60
	Site 4 ($n=4$)	8.50	175.10	5.50	31.00	6.60	27.30	7.80	3.00	21.30
	Site 5 ($n=5$)	6.30	181.20	4.50	25.90	5.10	28.90	8.30	15.60	19.80
	Site 6 ($n=4$)	7.40	178.10	5.30	28.40	5.80	28.10	8.50	9.30	20.50
	Site 7 ($n=5$)	5.90	179.40	4.70	27.30	5.30	31.40	8.60	17.30	19.60
	Site 8 ($n=4$)	7.20	184.80	4.20	23.10	4.80	27.70	7.90	12.10	20.10
	Site 9 ($n=5$)	6.50	182.10	4.40	25.20	5.30	29.50	8.20	14.70	19.80
UCC**		92.00	97.00	47.00	67.00	28.00	17.00	2.50	2.80	10.50
NASC*		125.00	130.00	58.00	85.00	45.00	20.00	3.00	3.20	12.00
TEL		52.30	50.00	15.90	124.00	18.70	30.20	2.00	3.00	3.00
PEL	> TEL	160.00	140.00	42.80	271.00	108.00	112.00	6.00	8.00	8.00
ERL	< TEL	81.00	85.00	20.90	150.00	34.00	46.70	3.00	4.00	4.00
ERM	> PEL	370.00	410.00	51.60	410.00	270.00	218.00	7.00	10.00	10.00

UCC upper continental crust (**Rudnick & Gao, 2003), *NASC* North American Shale Composite (*Taylor & McLennan, 1985), benchmarks: *TEL* threshold effect level, *PEL* probable effect level (MacDonald et al., 2000), *ERL* effect range—low, *ERM* effect range—median) (Long et al., 1995). A total of 82 samples were analyzed (41 per river). For each site, (n) indicates the number of individual sediment samples analyzed, and the values represent the mean concentration per site

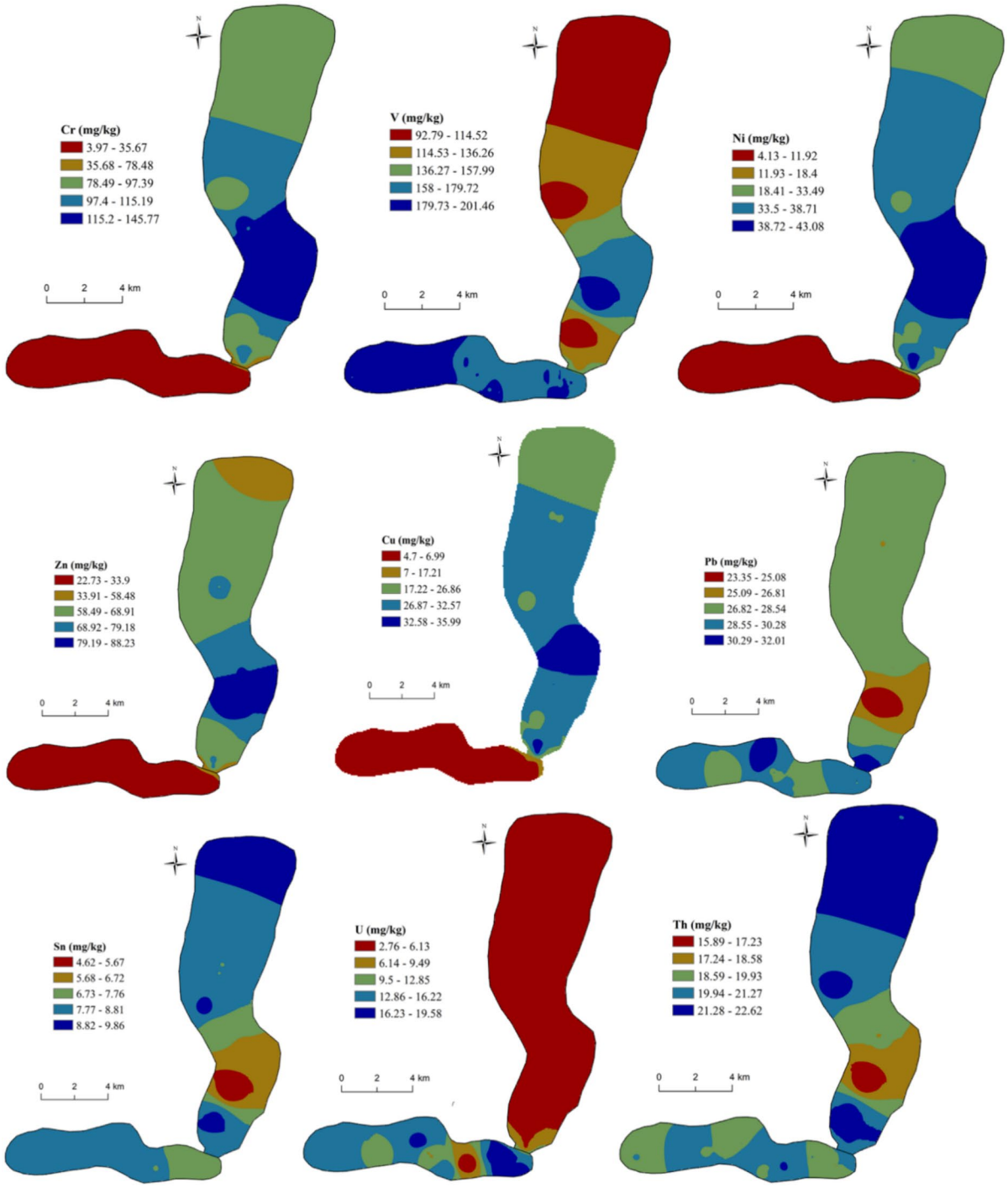


Fig. 2 Distribution of heavy metals along the Mbam and Noun River sediments

chemical weathering, leading to their consistent enrichment in sediments (Ambassa Bela et al., 2024). These results underscore the importance of considering both the geographical location of sampling sites and local hydrological characteristics when interpreting contamination dynamics in these fluvial systems.

Comparison with background values from Rudnick and Gao (2003) (Upper Continental Crust—UCC) and from Taylor and McLennan (1985) for the North American Shale Composite (NASC) reveals several major trends (Table 2). In the Mbam River, concentrations of Cr, Ni, Zn, Cu, Pb, and Sn generally exceed UCC values, reflecting both the geogenic contribution from the weathering of amphibolites and biotite gneisses enriched in Cr–Ni minerals, and localized anthropogenic inputs from artisanal mining, agricultural runoff, and urban effluents, particularly in downstream areas (Tiabou et al., 2025b). Cr and Ni levels occasionally exceed threshold effect level (TEL) thresholds (MacDonald et al., 2000) (Table 2), suggesting potential ecological effects at certain downstream stations. In the Noun River, V and U levels significantly exceed not only UCC and NASC values but also occasionally reach or surpass probable effect level (PEL) thresholds (MacDonald et al., 2000) (Table 2), suggesting a higher ecological risk for these specific metals.

Thorium (Th) concentrations remain relatively stable in both rivers, close to or slightly above geochemical background values, which likely reflects the natural contribution from the peralkaline granitic rocks and associated pegmatitic intrusions in the study area. These lithologies are known to be enriched in Th-bearing accessory minerals such as monazite, zircon, and allanite, whose weathering releases small but persistent amounts of thorium into river sediments without reaching sediment toxicity thresholds. Zn, Cu, and Pb are, on average, below effect range median (ERM) and probable effect level (PEL) thresholds (Long et al., 1995; MacDonald et al., 2000) (Table 2), indicating a low probability of severe biological effects, despite occasional localized enrichments. Spatial variation is particularly marked in the Mbam River, where sites near the mouth show the highest levels for most heavy metals, suggesting progressive downstream accumulation. In the Noun River, metal distribution is more uniform, but uranium and vanadium stand out with very high concentrations along the entire river before its confluence with the Mbam, indicating a dominant geogenic source or specific anthropogenic inputs to this basin. Comparisons with UCC, NASC, and ecotoxicological thresholds (TEL, PEL, ERL, ERM) (Long et al., 1995; MacDonald et al., 2000) show that some stations in both rivers display significant enrichments, particularly for V, Cr, Ni, and U (Table 2), justifying a detailed assessment of pollution and associated ecological risks in the following section.

Contamination levels

The assessment of sediment contamination levels in the Mbam and Noun Rivers is based on the combined analysis of four geochemical indices (Table 3, Fig. 3): the contamination factor (CF), the geo-accumulation index (Igeo), the enrichment factor (EF), and the overall Pollution Load Index (PLI). These indices were calculated using upper continental crust (UCC) background values from Rudnick and Gao (2003), with thorium (Th) used as the normalization element for EF.

The results reveal clear contrasts between the two river basins. In the Mbam River, CF values are generally greater than 1 for most metals (Table 3), indicating enrichment relative to natural background levels. Several sites show CF values above 3 for Cr, Ni, V, and Cu (Fig. 3a), particularly in downstream stations (Sites 1 to 3), corresponding to considerable contamination. Conversely, upstream sites (Sites 8 and 9) display lower CF values, often below 2 (Table 3, Fig. 3a), confirming a decrease in contaminant load upstream. EF values confirm this trend, with moderate to considerable enrichments (EF between 1.5 and 5) for Sn, Cr, Cu, V, and Pb in Mbam sediments (Fig. 3b). Notably, Sn shows high EF values at nearly all Mbam sites, suggesting anthropogenic contributions linked to local socio-economic activities. In the lower Mbam basin, several artisanal and small-scale workshops dealing with metal scraps, vehicle repair, and battery recycling are established along the riverbanks. These activities, together with domestic waste disposal and the use of tin-based materials in villages such as Deuk, Dioma, and Mantoum, likely contribute to the elevated Sn concentrations observed in downstream sediments. Occasional agricultural runoff containing metallic residues from fertilizers and tools may also enhance local enrichment. In contrast, EF values for Ni and U generally remain below 1.5 (Fig. 3b), suggesting a predominantly natural origin for these elements in this river.

In the Noun River, CF values for traditional heavy metals (Cr, Ni, Zn, Cu, Pb) are generally below 1

Table 3 Pollution indices of metal contamination (CF, Igeo, EF, PLI) and ecological risk indices in sediments from the Mbam and Noun Rivers

Rivers	Mbam									Noun								
Sites (samples)	Site 1 ($n=4$)	Site 2 ($n=5$)	Site 3 ($n=4$)	Site 4 ($n=5$)	Site 5 ($n=4$)	Site 6 ($n=4$)	Site 7 ($n=5$)	Site 8 ($n=5$)	Site 9 ($n=5$)	Site 1 ($n=5$)	Site 2 ($n=5$)	Site 3 ($n=4$)	Site 4 ($n=4$)	Site 5 ($n=5$)	Site 6 ($n=4$)	Site 7 ($n=5$)	Site 8 ($n=4$)	Site 9 ($n=5$)
CF_Cr	1.19	0.97	1.55	1.37	1.27	0.97	1.17	0.96	0.98	0.04	0.09	0.06	0.09	0.07	0.08	0.06	0.08	0.07
CF_V	1.29	1.02	2.04	1.66	1.50	1.02	1.34	1.06	0.98	1.84	1.86	1.85	1.81	1.87	1.84	1.85	1.91	1.88
CF_Ni	0.91	0.69	0.89	0.90	0.85	0.69	0.80	0.73	0.66	0.10	0.10	0.10	0.12	0.10	0.11	0.10	0.09	0.09
CF_Zn	1.06	0.90	1.29	1.18	1.11	0.90	1.04	0.97	0.83	0.42	0.40	0.41	0.46	0.39	0.42	0.41	0.34	0.38
CF_Cu	1.26	0.93	1.14	1.20	1.14	0.93	1.07	0.96	0.91	0.19	0.19	0.19	0.24	0.18	0.21	0.19	0.17	0.19
CF_Pb	1.82	1.62	1.40	1.61	1.61	1.62	1.61	1.59	1.65	1.78	1.68	1.72	1.61	1.70	1.65	1.85	1.63	1.74
CF_Sn	3.44	3.64	1.88	2.64	2.88	3.60	3.12	3.32	3.88	2.84	2.76	2.96	3.12	3.32	3.40	3.44	3.16	3.28
CF_U	1.96	1.61	1.00	1.46	1.50	1.63	1.54	1.64	1.61	5.50	6.89	6.18	1.07	5.57	3.32	6.18	4.32	5.25
CF_Th	2.02	2.08	1.54	1.78	1.85	2.08	1.93	2.11	2.04	1.96	1.78	1.87	2.03	1.89	1.95	1.87	1.91	1.89
Igeo_Cr	−0.34	−0.63	0.05	−0.13	−0.24	−0.62	−0.36	−0.65	−0.61	−5.15	−4.13	−4.55	−4.02	−4.45	−4.22	−4.55	−4.26	−4.41
Igeo_V	−0.22	−0.56	0.45	0.15	0.00	−0.56	−0.16	−0.51	−0.62	0.29	0.31	0.30	0.27	0.32	0.29	0.30	0.34	0.32
Igeo_Ni	−0.73	−1.11	−0.75	−0.74	−0.82	−1.11	−0.91	−1.04	−1.19	−3.88	−3.94	−3.91	−3.68	−3.97	−3.73	−3.91	−4.07	−4.00
Igeo_Zn	−0.50	−0.74	−0.21	−0.35	−0.44	−0.74	−0.53	−0.63	−0.85	−1.84	−1.92	−1.88	−1.70	−1.96	−1.82	−1.88	−2.12	−2.00
Igeo_Cu	−0.25	−0.69	−0.39	−0.32	−0.40	−0.69	−0.49	−0.65	−0.73	−2.96	−2.99	−2.99	−2.67	−3.04	−2.86	−2.99	−3.13	−2.99
Igeo_Pb	0.28	0.11	−0.10	0.10	0.10	0.11	0.11	0.09	0.14	0.24	0.16	0.20	0.10	0.18	0.14	0.30	0.12	0.21
Igeo_Sn	1.20	1.28	0.33	0.82	0.94	1.26	1.06	1.15	1.37	0.92	0.88	0.98	1.06	1.15	1.18	1.20	1.07	1.13
Igeo_U	0.39	0.10	−0.58	−0.03	0.00	0.12	0.04	0.13	0.10	1.87	2.20	2.04	−0.49	1.89	1.15	2.04	1.53	1.81
Igeo_Th	0.43	0.47	0.04	0.25	0.30	0.47	0.36	0.50	0.44	0.39	0.25	0.32	0.44	0.33	0.38	0.32	0.35	0.33
EF_Cr	0.59	0.47	1.01	0.77	0.69	0.47	0.61	0.45	0.48	0.02	0.05	0.03	0.05	0.04	0.04	0.03	0.04	0.04
EF_V	0.64	0.49	1.33	0.93	0.81	0.49	0.69	0.50	0.48	0.94	1.05	0.99	0.89	0.99	0.94	0.99	1.00	1.00
EF_Ni	0.45	0.33	0.58	0.51	0.46	0.33	0.41	0.35	0.32	0.05	0.05	0.05	0.06	0.05	0.06	0.05	0.05	0.05
EF_Zn	0.53	0.43	0.84	0.66	0.60	0.43	0.54	0.46	0.41	0.21	0.22	0.22	0.23	0.20	0.22	0.22	0.18	0.20
EF_Cu	0.63	0.45	0.74	0.68	0.61	0.45	0.55	0.45	0.45	0.10	0.11	0.10	0.12	0.10	0.11	0.10	0.09	0.10
EF_Pb	0.90	0.78	0.91	0.90	0.87	0.78	0.84	0.75	0.81	0.91	0.94	0.92	0.79	0.90	0.85	0.99	0.85	0.92
EF_Sn	1.70	1.75	1.22	1.48	1.56	1.73	1.62	1.57	1.90	1.45	1.55	1.59	1.54	1.76	1.74	1.84	1.65	1.74

Table 3 (continued)

Rivers	Mbam									Noun								
Sites (samples)	Site 1 (n=4)	Site 2 (n=5)	Site 3 (n=4)	Site 4 (n=5)	Site 5 (n=4)	Site 6 (n=4)	Site 7 (n=5)	Site 8 (n=5)	Site 9 (n=5)	Site 1 (n=5)	Site 2 (n=5)	Site 3 (n=4)	Site 4 (n=4)	Site 5 (n=5)	Site 6 (n=4)	Site 7 (n=5)	Site 8 (n=4)	Site 9 (n=5)
EF_U	0.97	0.77	0.65	0.82	0.81	0.78	0.80	0.78	0.79	2.80	3.87	3.31	0.53	2.95	1.70	3.31	2.26	2.78
PLI	1.53	1.31	1.37	1.47	1.44	1.31	1.40	1.33	1.29	0.69	0.74	0.72	0.66	0.72	0.72	0.74	0.69	0.72
Er_Cr	2.37	1.94	3.11	2.74	2.54	1.95	2.34	1.91	1.96	0.08	0.17	0.13	0.18	0.14	0.16	0.13	0.16	0.14
Er_V	2.57	2.03	4.09	3.33	3.00	2.03	2.68	2.11	1.95	3.68	3.72	3.70	3.61	3.74	3.67	3.70	3.81	3.75
Er_Ni	4.53	3.46	4.47	4.50	4.23	3.47	3.98	3.66	3.29	0.51	0.49	0.50	0.59	0.48	0.56	0.50	0.45	0.47
Er_Zn	1.06	0.89	1.29	1.18	1.11	0.90	1.04	0.97	0.83	0.42	0.40	0.41	0.46	0.39	0.42	0.41	0.34	0.38
Er_Cu	6.32	4.66	5.71	6.02	5.68	4.66	5.34	4.79	4.54	0.96	0.95	0.95	1.18	0.91	1.04	0.95	0.86	0.95
Er_Pb	9.08	8.11	7.00	8.03	8.06	8.12	8.07	7.97	8.26	8.88	8.38	8.62	8.03	8.50	8.26	9.24	8.15	8.68
Er_Sn	17.20	18.20	9.40	13.20	14.40	18.00	15.60	16.60	19.40	14.20	13.80	14.80	15.60	16.60	17.00	17.20	15.80	16.40
Er_U	58.92	48.21	30.00	43.93	45.00	48.75	46.34	49.29	48.21	165.00	206.79	185.36	32.14	167.14	99.64	185.36	129.64	157.50
Er_Th	10.10	10.38	7.71	8.90	9.24	10.38	9.64	10.57	10.19	9.81	8.90	9.33	10.14	9.43	9.76	9.33	9.57	9.43
RI	112.18	97.91	72.79	91.83	93.26	98.25	95.04	97.87	98.64	203.55	243.60	223.79	71.94	207.32	140.53	226.81	168.78	197.69

(Table 3), indicating low contamination relative to background levels. However, CF and EF values for vanadium (V) and uranium (U) are notably higher at all sites. CF values for V often exceed 1.8, and U reaches values up to 5.5 (Table 3, Fig. 3a'). EF values for V in the Noun are close to or greater than 1, while U values frequently exceed the threshold value of 2.8 (Table 3, Fig. 3b'), indicating significant enrichment that may reflect a geogenic origin (naturally enriched formation) or specific anthropogenic sources in this basin.

The Igeo index mirrors these observations: in the Mbam, Igeo values for Cr, Ni, V, Cu, and Sn frequently range between 0 and 2 (Table 3, Fig. 3c'), indicating low to moderate pollution. Some sites display Igeo values above 2 for Cr and Cu (Fig. 3), signaling moderately strong pollution in downstream sectors. In the Noun, only V and U occasionally reach Igeo > 1, while other metals show Igeo values below 0 (Table 3, Fig. 3c'), indicating no pollution according to this index. The Pollution Load Index (PLI), calculated as the geometric mean of CF values for all metals, provides an overall site-level contamination assessment. PLI values in the Mbam range between 1.29 and 1.53 (Table 3, Fig. 3a), indicating widespread but moderate pollution. In contrast, sites in the Noun present PLI values well below 1 (between 0.66 and 0.74), reflecting low overall contamination pressure despite localized enrichments in V and U (Fig. 3a').

These results confirm that the Mbam River is significantly more contaminated than the Noun for most heavy metals, particularly in its downstream section. The use of Th as a reference element allowed for distinguishing truly significant enrichments of either anthropogenic or natural origin. This distinction is critical for guiding environmental monitoring efforts and watershed management strategies.

Ecological risks

The assessment of the potential ecological risk posed by heavy metals in sediments from the Mbam and Noun Rivers is based on the calculation of the individual risk factor (*Er*) and the overall ecological risk index (*RI*), following the methodology of Hakanson (1980). These indices combine the contamination level (CF) with the relative toxicity of each metal,

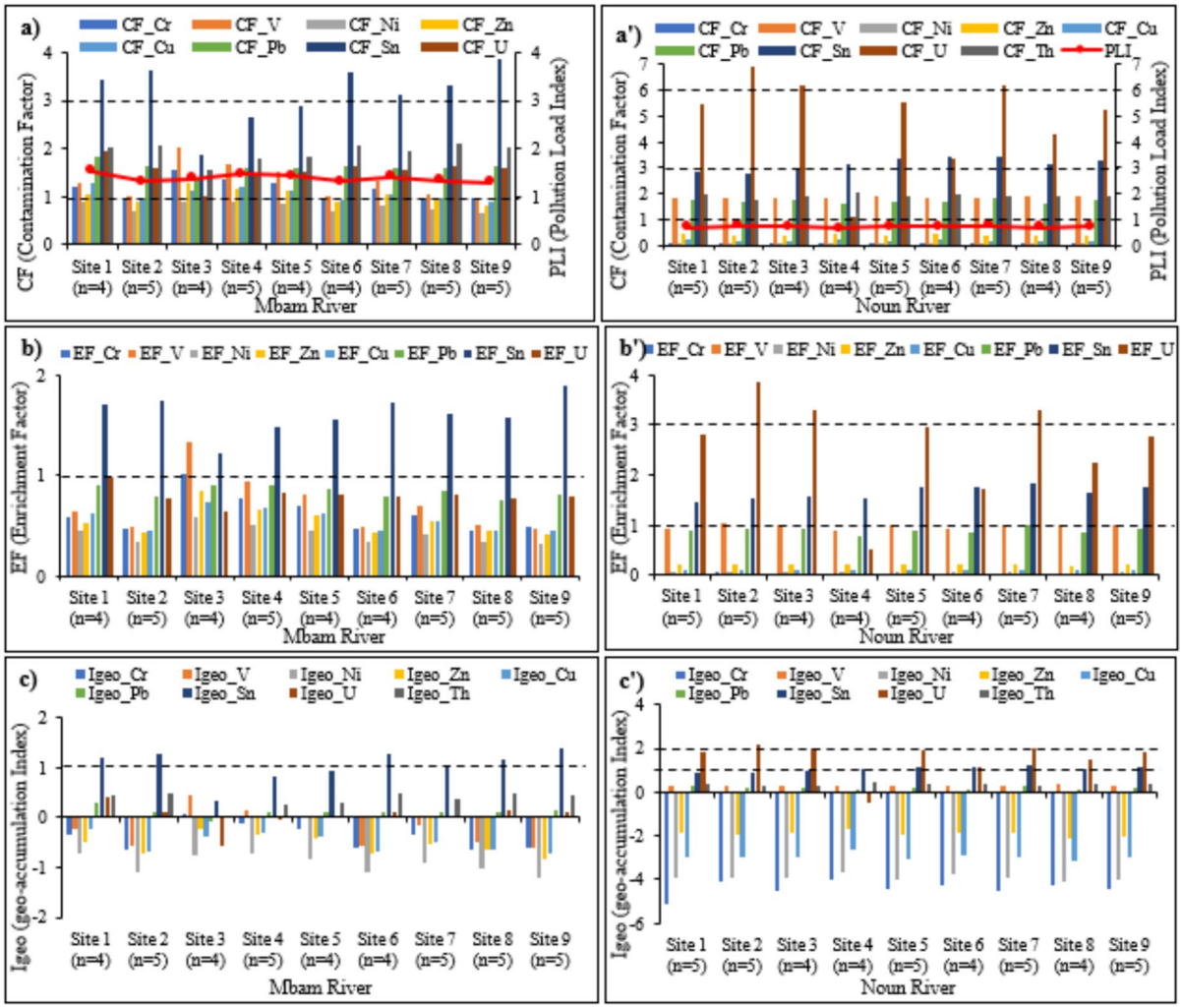


Fig. 3 Calculation indices of **a** and **a'** contamination factor (CF) and pollution load index (PLI) for Mbam and Noun Rivers, respectively. **b** and **b'** enrichment factor (EF) for Mbam and Noun Rivers, respectively. **c** and **c'** Geo-accumulation index (Igeo) for Mbam and Noun Rivers, respectively

providing a quantitative estimate of the ecological threat posed by contaminated sediments.

Individual risk factors (Er)

The results show that in the Mbam River, the metals with the highest *Er* values are primarily uranium (U), followed by lead (Pb) and tin (Sn). $Er_{(U)}$ values range from 30 to nearly 59, placing uranium's contribution in the category of moderate to high ecological risk (Table 3, Fig. 4a). Lead and tin generally display *Er* values between 7 and 19, corresponding to low to moderate risk, while other metals (Cr, Ni, Cu, V, Zn) have low *Er* values (< 10) (Table 3, Fig. 4a). Overall, most heavy metals in the Mbam contribute to a low to moderate ecological risk, with uranium exerting the most significant influence.

In the Noun River, the results are more concerning. $Er_{(U)}$ values reach extremely high levels, exceeding 200 (Table 3, Fig. 4b) at some sites (maximum≈206). According to Hakanson's (1980) classification, these values indicate a very high ecological risk. Vanadium (V) also contributes notably, with $Er_{(V)}$ values close to 3.7, while lead and tin exhibit *Er* values similar to those observed in the Mbam (Table 3, Fig. 4a).

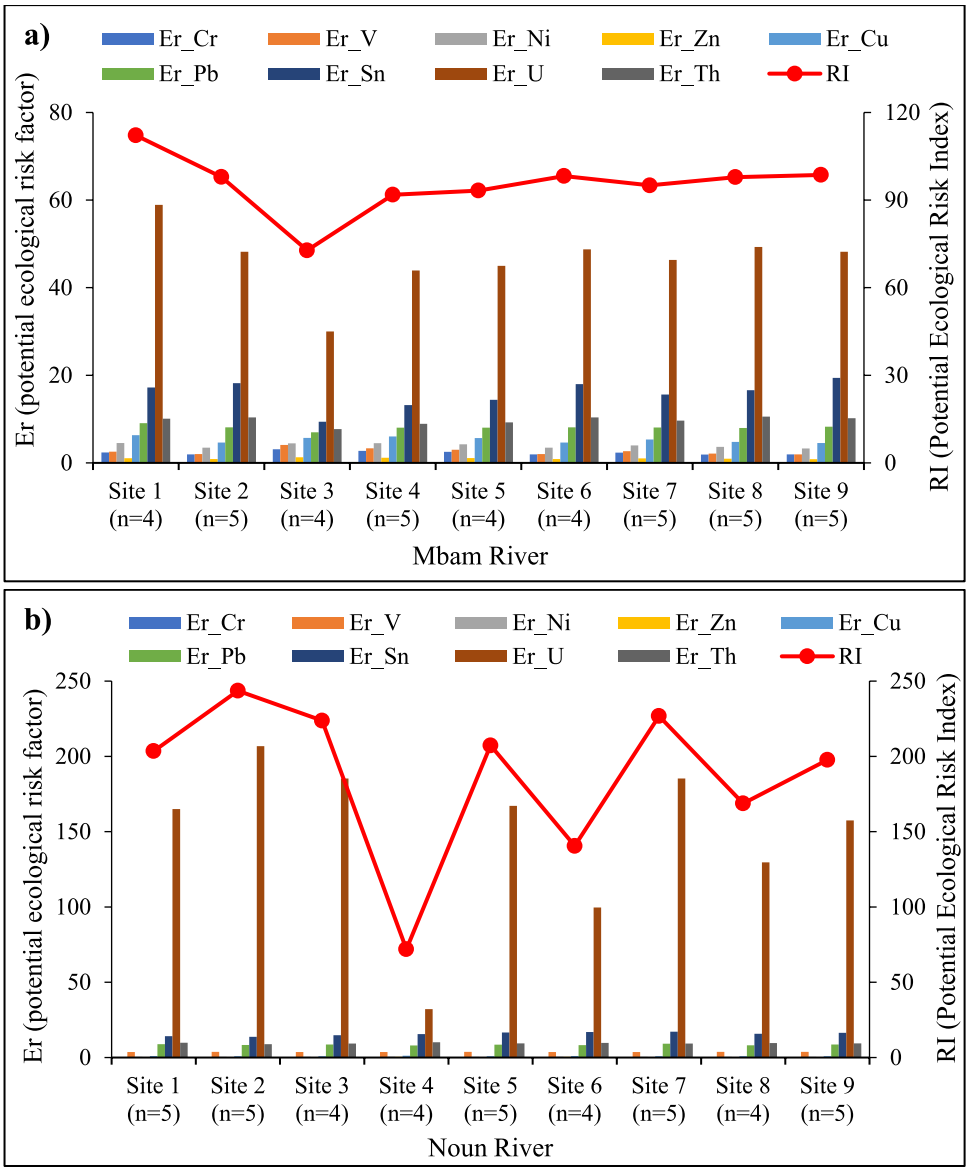


Fig. 4. Indices of **a)** potential ecological factor (*Eri*); and **b)** potential ecological index (RI) in sediments from Mbam and Noun Rivers

Fig. 4 Indices of **a** potential ecological factor (*Eri*) and **b** potential ecological index (RI) in sediments from Mbam and Noun Rivers

Other heavy metals, although weakly contaminated, add only marginal contributions. This strong uranium presence, combined with vanadium enrichment, makes the Noun basin more ecologically vulnerable than the Mbam.

Ecological risk index (RI)

The overall *RI*, obtained by summing the *Er* values of all metals for each site, provides an assessment of the combined impact of contamination. In the Mbam

River, *RI* values range between 72 and 112 (Table 3, Fig. 4a), corresponding to a moderate overall ecological risk according to the classification in Table 1. In the Noun River, *RI* values are significantly higher, ranging from 71 to over 243, with several sites exceeding the threshold of 150 (Table 3, Fig. 4b). This places these sites in the considerable ecological risk category, with some points approaching a very high risk level (> 200) (Table 3, Fig. 4). This marked difference between the two rivers highlights the dominant role of uranium and vanadium as the main contributors to ecological degradation in the Noun.

Mbam vs. Noun comparison

The comparison between the two basins shows that the Mbam River is characterized by moderate multi-metal contamination, with a relatively contained overall ecological risk, as *RI* values generally remain below 150 despite occasional uranium enrichments. In contrast, the Noun River presents a far more concerning situation, marked by extremely high uranium levels that result in individual risk factors ($Er_{(U)}$) and overall indices (*RI*) well above those observed in the Mbam. This ecological risk disparity appears to stem from both natural geochemical differences between the two basins and potential anthropogenic sources. The evaluation of ecological risk indices reveals that although both rivers display notable levels of heavy metal contamination, the Noun poses the greater threat to the aquatic environment, primarily due to its exceptional enrichment in uranium and vanadium.

Human health risk assessment

The potential human health impact of sediment-bound metals in the Mbam and Noun Rivers was evaluated using two sets of indicators: (i) non-carcinogenic risk indices, such as the hazard index (HI), and (ii) carcinogenic risk indices, such as the total carcinogenic risk (TCR). These indices were calculated separately for children and adults at each sampling site, based on chronic incidental ingestion of sediments following the USEPA exposure model (Li et al., 2014; Shao et al., 2021; USEPA, 1989, 2011). The main results are summarized in Tables 4 and S1.

Table 4 Carcinogenic risk (CR) of chromium [Cr (VI)], nickel (Ni), and uranium (U) and total carcinogenic risk (TCR) in surface sediments from the Mbam and Noun Rivers

River	Site	CR_Cr (VI)	CR_Ni	CR_U	TCR
Mbam	Site 1	5.98e-05	3.92e-05	9.04e-11	9.91e-05
	Site 2	4.89e-05	3.00e-05	7.40e-11	7.89e-05
	Site 3	7.84e-05	3.87e-05	4.60e-11	1.17e-04
	Site 4	6.91e-05	3.89e-05	6.74e-11	1.08e-04
	Site 5	6.41e-05	3.66e-05	6.90e-11	1.01e-04
	Site 6	4.90e-05	3.00e-05	7.48e-11	7.91e-05
	Site 7	5.91e-05	3.45e-05	7.10e-11	9.35e-05
	Site 8	4.82e-05	3.17e-05	7.56e-11	7.98e-05
	Site 9	4.94e-05	2.84e-05	7.40e-11	7.79e-05
Noun	Site 1	2.14e-06	4.42e-06	2.53e-10	6.56e-06
	Site 2	4.33e-06	4.23e-06	3.17e-10	8.56e-06
	Site 3	3.23e-06	4.32e-06	2.84e-10	7.55e-06
	Site 4	4.66e-06	5.06e-06	4.93e-11	9.72e-06
	Site 5	3.45e-06	4.14e-06	2.56e-10	7.59e-06
	Site 6	4.05e-06	4.88e-06	1.53e-10	8.93e-06
	Site 7	3.23e-06	4.33e-06	2.84e-10	7.56e-06
	Site 8	3.95e-06	3.87e-06	1.99e-10	7.81e-06
	Site 9	3.56e-06	4.05e-06	2.42e-10	7.61e-06

Non-carcinogenic risk (HI)

Non-carcinogenic risk was assessed through the hazard index (HI), which represents the sum of the individual hazard quotients (HQ) for each metal (Cr, V, Ni, Zn, Cu, Sn, U) (Table S1). HI values below 1 indicate that adverse health effects are unlikely under the assumed exposure conditions, whereas $HI > 1$ suggests a potential for non-carcinogenic effects (USEPA, 2011).

In the Mbam River, HI values for children range from 0.192 to 0.336 across sites, with the highest values found at Sites 3 and 4 (HI_child = 0.336 and 0.289, respectively) (Table S1). These downstream stations correspond to areas where overall metal contamination is also elevated (high CF and PLI, Table 3), suggesting that multi-metal exposure is most critical in the lower Mbam. Upstream sites (Sites 8 and 9) show lower HI_child values (≈0.19–0.22) (Table S1), consistent with a general decrease in metal concentrations in the upper reaches of the river (see section "Heavy metal concentrations"). For adults, HI values are systematically lower, between 0.021 and 0.036 (Table S1), reflecting

lower ingestion rates relative to body weight. All HI values in the Mbam remain well below 1, indicating no significant non-carcinogenic risk for either adults or children at the time of sampling.

In the Noun River, HI_child values are slightly higher and display less spatial variability, ranging from 0.300 to 0.339 (Table S1). These values are consistently above those observed at most Mbam sites, indicating a higher potential exposure risk for children along the Noun. This pattern reflects the persistent enrichment in uranium (U) and vanadium (V) observed throughout the Noun basin (section "Heavy metal concentrations"), which strongly contributes to the cumulative HI. HI_adult values in the Noun are also low (0.030–0.034), comparable to the Mbam. Importantly, no site in either river shows $HI \geq 1$ (Table S1). This indicates that under chronic exposure scenarios, non-carcinogenic health effects are not expected for local populations, although the systematically higher HI values calculated for children suggest that they represent the more vulnerable group.

Overall, the spatial distribution of HI values indicates two key points: (i) The downstream Mbam shows localized elevation of non-carcinogenic risk indices, likely linked to cumulative multi-metal inputs (Cr, Ni, Cu, Sn), whereas (ii) the Noun shows uniformly elevated HI_child values at all sites, reflecting persistent U and V enrichment across the entire basin rather than a localized hotspot.

Carcinogenic risk (TCR)

Carcinogenic risk was estimated using the total carcinogenic risk (TCR), which sums the individual carcinogenic risks (CR) associated with chromium (assumed as Cr (VI) for conservative screening), nickel (Ni), and uranium (U). TCR is interpreted as the probability of developing cancer over a lifetime due to long-term exposure. According to USEPA guidelines, a TCR between 10^{-6} and 10^{-4} is considered within an acceptable risk range (USEPA, 1989, 2011; Wang et al., 2020; Wu et al., 2022). Values higher than 10^{-4} would be of concern.

In the Mbam River, TCR_child values range from approximately 7.6×10^{-5} to 1.17×10^{-4} (Table 4). The highest values occur at Sites 3 and 4 in the downstream sector, where metal loads are also highest. These TCR_child values approach the upper part of the generally accepted risk range (10^{-6}–10^{-4}), indicating a non-negligible long-term carcinogenic risk for children at the most impacted sites in the lower Mbam. TCR_adult values in the Mbam range from 3.27×10^{-5} to 5.02×10^{-5}, i.e., lower than TCR_child but still on the order of 10^{-5}, which remains within the acceptable risk range.

In the Noun River, TCR_child values are systematically lower, between 4.20×10^{-6} and 1.63×10^{-5}, and TCR_adult values range from 1.79×10^{-6} to 6.98×10^{-6} (Table 4). These values fall in the lower part of the acceptable range. The relatively low TCR in the Noun, compared to the Mbam, indicates that although uranium and vanadium are highly enriched in Noun sediments (see section "Heavy metal concentrations"), their contribution to modelled lifetime carcinogenic risk remains smaller than the combined effect of Cr and Ni observed in the Mbam.

Taken together, these results indicate that (i) all sites in both rivers remain within the generally accepted carcinogenic risk range defined by regulatory criteria, but (ii) the downstream Mbam (Sites 1–4) presents the highest TCR values, particularly for children, and therefore represents the sector of greatest concern from a human health perspective.

Implications for exposure and management

The combined interpretation of HI and TCR suggests that the current levels of sediment-associated metals do not indicate an immediate toxicological threat for local populations under typical exposure assumptions. However, children systematically display higher HI and TCR values than adults at every site, underscoring their higher vulnerability to chronic exposure through incidental ingestion of riverbank sediments. This is particularly relevant in communities where river sediments are handled directly for domestic, artisanal, or agricultural activities. From a spatial standpoint, the Mbam River exhibits a localized but higher potential carcinogenic risk in its downstream sector, whereas the Noun River shows a more homogeneous increase in non-carcinogenic risk indices linked to uranium and vanadium. These results support the need for targeted monitoring (i) in the lower Mbam, where multi-metal exposure (notably Cr, Ni, and Cu) elevates both HI and TCR, and (ii)

throughout the Noun basin, where natural or mixed geogenic/anthropogenic inputs of U and V increase long-term exposure pressure for resident populations. These findings provide an essential health-based complement to the contamination (section "Contamination levels") and ecological risk assessment (section "Ecological risks") and justify the integration of human health criteria into future watershed management and land-use planning in the Mbam–Noun system.

Statistical analyses

Multivariate statistical analysis, including correlation matrix, principal component analysis (PCA), and hierarchical clustering, was used to explore

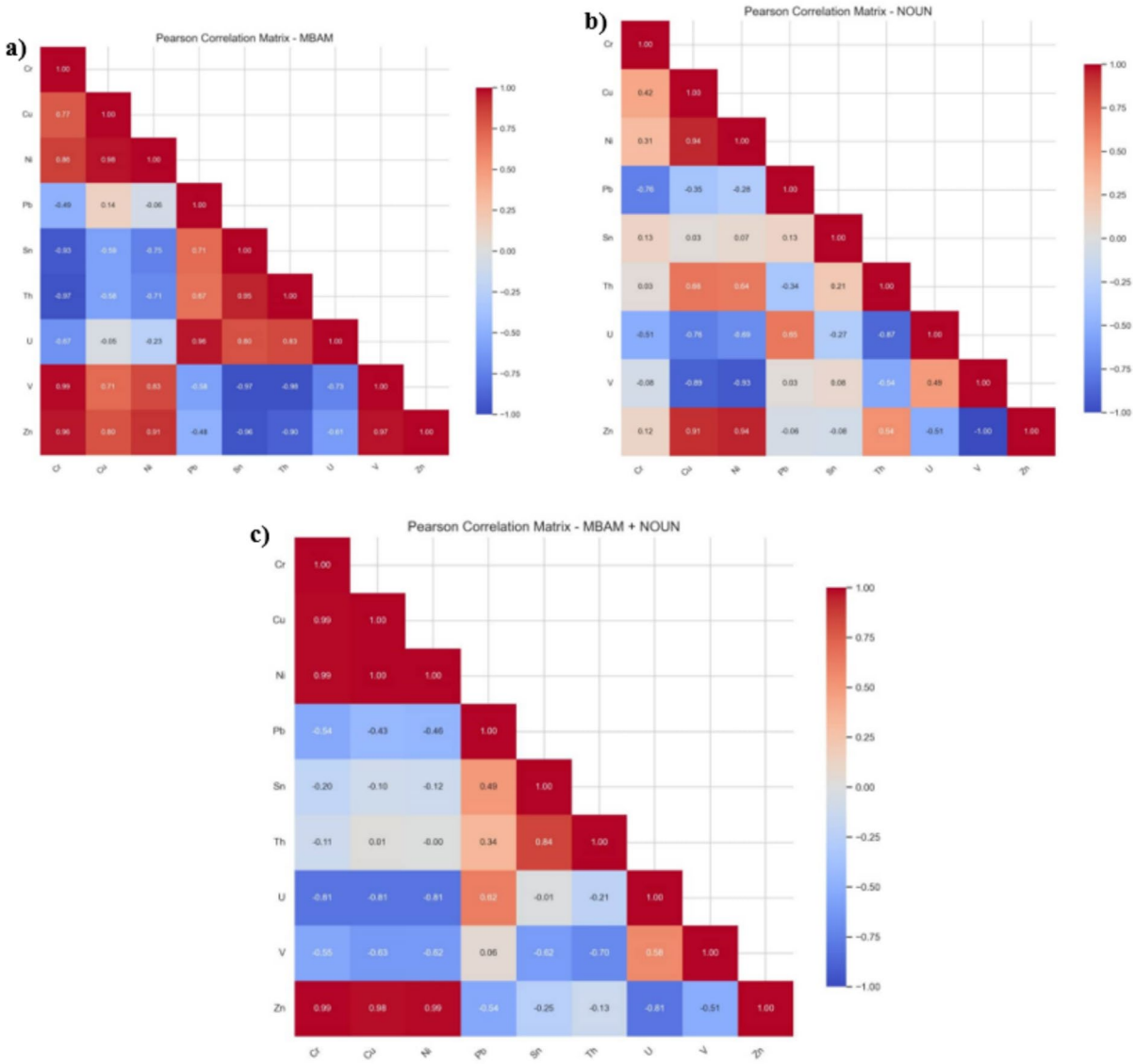


Fig. 5 Pearson correlation matrix of heavy metals along the river sediments. **a** Mbam, **b** Noun, and **c** Mbam + Noun

relationships among heavy metals and to distinguish between sampling sites in the Mbam and Noun Rivers.

The correlation matrix revealed significant relationships between several metals (Fig. 5a, b, c). In the Mbam, chromium (Cr), zinc (Zn), copper (Cu), and nickel (Ni) exhibited moderate to strong positive correlations, suggesting a common origin or similar geochemical processes (Fig. 5a), likely related to the weathering of local lithological formations or combined anthropogenic inputs (Tiabou et al., 2025c; Sigue et al., 2025b). In the Noun, correlations were more scattered; uranium (U) and vanadium (V) were less correlated with other

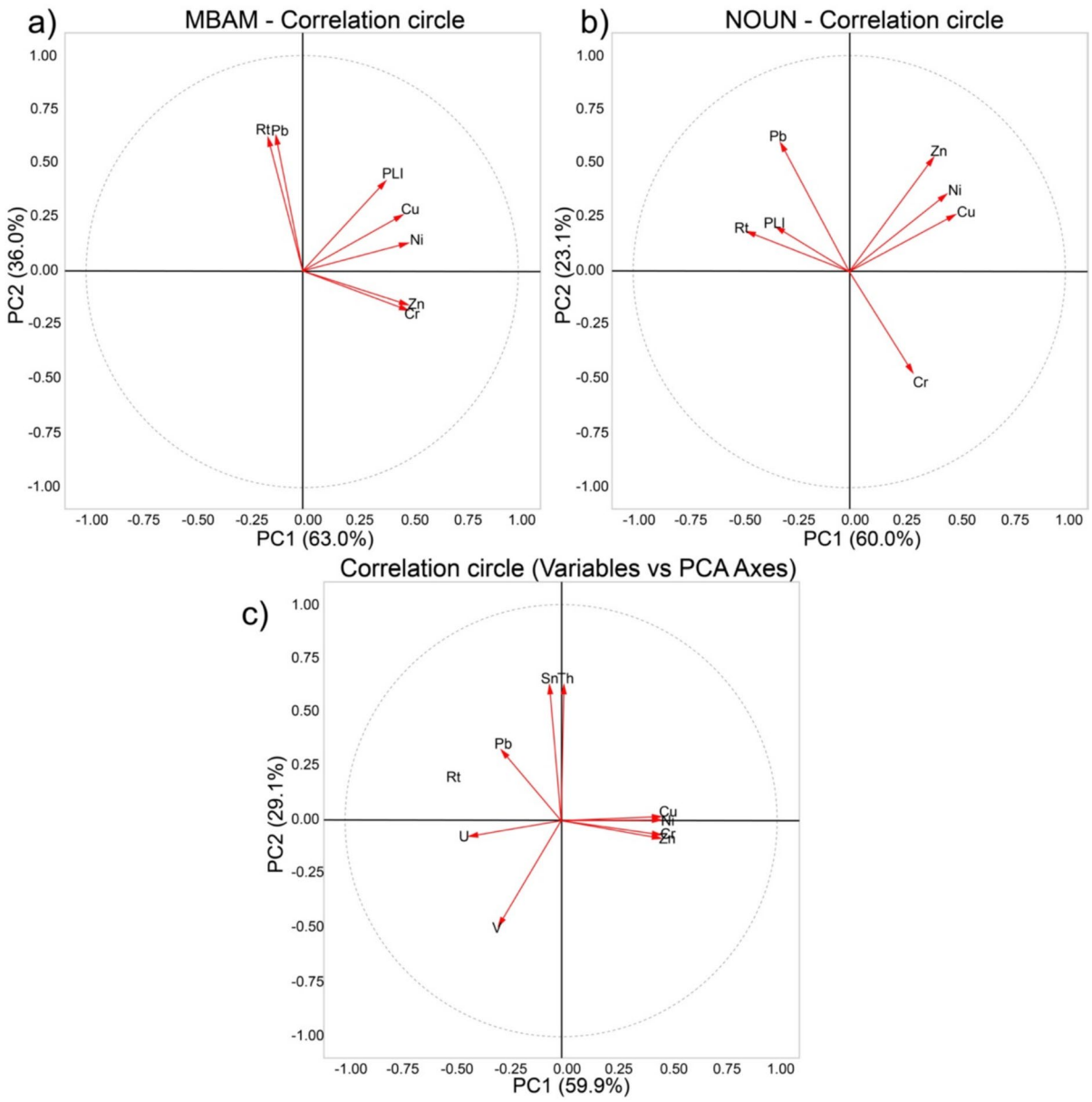
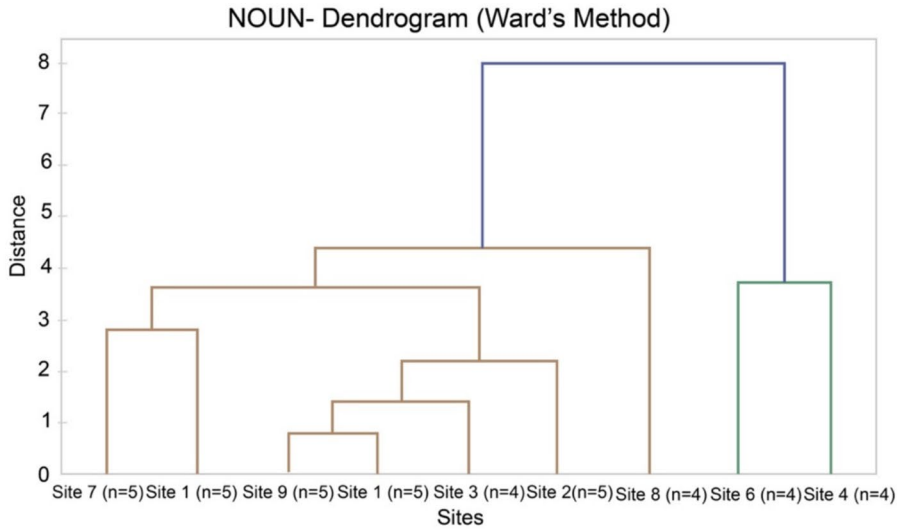
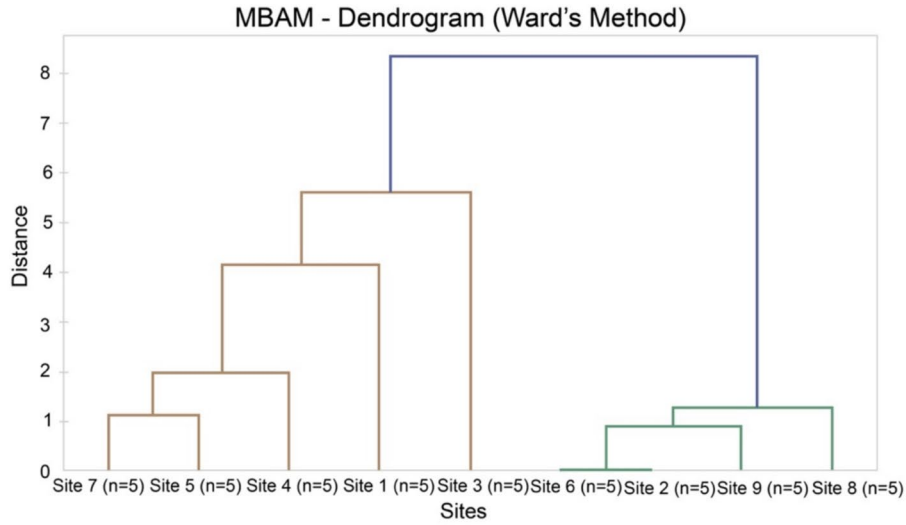
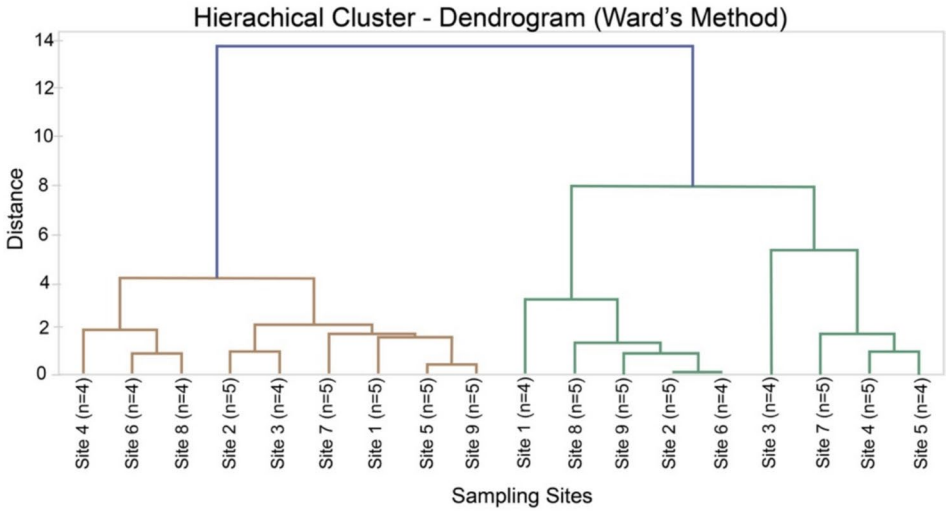


Fig. 6 Principal component analysis of studied variables (trace metals, CF, EF, Igeo, PLI, and RI) of river sediments from **a** Mbam, **b** Noun, and **c** Mbam + Noun


NOUN- Dendrogram (Ward's Method)
Distance
0
1
2
3
4
5
6
7
8
Site 7 (n=5)
Site 1 (n=5)
Site 9 (n=5)
Site 1 (n=5)
Site 3 (n=4)
Site 2(n=5)
Site 8 (n=4)
Site 6 (n=4)
Site 4 (n=4)
Sites


MBAM - Dendrogram (Ward's Method)
Distance
0
1
2
3
4
5
6
7
8
Site 7 (n=5)
Site 5 (n=5)
Site 4 (n=5)
Site 1 (n=5)
Site 3 (n=5)
Site 6 (n=5)
Site 2 (n=5)
Site 9 (n=5)
Site 8 (n=5)
Sites


Hierachical Cluster - Dendrogram (Ward's Method)
Distance
0
2
4
6
8
10
12
14
Site 4 (n=4)
Site 6 (n=4)
Site 8 (n=4)
Site 2 (n=5)
Site 3 (n=4)
Site 7 (n=5)
Site 1 (n=5)
Site 5 (n=5)
Site 9 (n=5)
Site 1 (n=4)
Site 8 (n=5)
Site 9 (n=5)
Site 2 (n=5)
Site 6 (n=4)
Site 3 (n=4)
Site 7 (n=5)
Site 4 (n=5)
Site 5 (n=4)
Sampling Sites

◂**Fig. 7** Dendrograms of cluster analysis of the metal concentrations analyzed along the river sediments of **a** Mbam, **b** Noun, and **c** Mbam + Noun

metals (Fig. 5b), confirming their specific enrichment distinct from other contaminants.

Principal component analysis reduced data dimensionality and identified the principal components explaining most of the variance (Fig. 6). For the Mbam, the first two axes (PC1 and PC2) explained approximately 63 and 36% of the total variance, respectively, accounting for nearly 99% in total (Fig. 6a). PC1 was strongly loaded with Cr, Zn, Ni, and Cu, indicating that these metals covary and likely originate from common geochemical sources. PC2 was primarily influenced by lead (Pb) and the pollution load index (PLI) (Fig. 6a), suggesting a specific Pb contribution, possibly linked to local human activities such as mining or urban discharges. In the Noun, the first two components ($PC1 = 60\%$, $PC2 = 23\%$) explained about 83% of the variance (Fig. 6b). PC1 grouped Cr, Zn, Ni, and Cu, while PC2 highlighted Pb and ecological risk indices (RI and PLI) (Fig. 6b). These results indicate that in the Noun, classical heavy metals coevolve, but Pb and global contamination indicators follow a distinct pattern. In the combined analysis of both rivers, PCA ($PC1 = 59.9\%$, $PC2 = 29.1\%$) explained about 89% of the total variance (Fig. 6c). PC1 was mainly associated with Cr, Zn, Cu, Ni, and the enrichment factor (EF), reflecting the combined effect of multi-metal contamination, while PC2 was loaded with U, V, Th, and Sn (Fig. 6c), underscoring their specificity in the geochemical signature of certain sites, particularly in the Noun.

Hierarchical clustering (dendrograms) separated sites into distinct clusters. In the Mbam, downstream sites (1 to 4) were grouped, reflecting similar contamination patterns, while upstream sites (8 and 9) formed a separate cluster with lower heavy metal levels (Fig. 7a). In the Noun, two main clusters also emerged: one group of sites with low and homogeneous contamination for most metals and another with pronounced enrichment in uranium and vanadium (Fig. 7b). The combined analysis of both rivers confirmed this separation, grouping Noun sites heavily contaminated with uranium into a cluster distinct from Mbam sites (Fig. 7c).

Overall, these statistical analyses highlight a differentiated spatial and chemical structure of sediments. They show that although multi-metal contamination (Cr, Ni, Zn, Cu) is common to both rivers, the specific contributions of uranium, vanadium, and certain other heavy elements clearly distinguish sites, particularly in the Noun. These results confirm the need for a targeted environmental management approach that accounts for distinct geochemical and anthropogenic sources in each basin.

Discussion

Potential origin of heavy metals

The elevated levels of heavy metals such as Cr, V, Ni, Zn, Cu, Pb, Sn, U, and Th observed in the sediments of the Mbam and Noun Rivers result from a combination of natural geogenic sources and anthropogenic inputs, mainly originating from agricultural runoff enriched with fertilizers and pesticides, domestic wastewater discharges, and surface erosion from nearby settlements. On the one hand, the local lithology, dominated by Precambrian basement rocks rich in gneiss, amphibolites, and granitoids, as well as volcanic formations from the Western Highlands and the Cameroon Volcanic Line, constitutes a major source of metals such as Cr, Ni, V, and U through the erosion and weathering of parent rocks (Fig. 1b). This pattern is consistent with observations from the Lom watershed in eastern Cameroon (Edjengté Doumo et al., 2023; Yiika et al., 2023; NdemaMbongué et al., 2024) and the Lékié River, where geological weathering plays a key role in the enrichment of certain metals (Ekoa Bessa et al., 2024).

On the other hand, the socio-economic context of the studied basins reveals several activities likely to generate anthropogenic inputs. Although urbanization is relatively limited, the presence of small-scale urban trades such as automobile repair shops, mechanical workshops, and informal handling of metallic waste can contribute to the release of metals like Pb, Cu, and Sn. These dynamics are consistent with findings from peri-urban areas of Yaoundé, where emissions from garages and urban agriculture practices involving pesticide- or fertilizer-enriched inputs have been

associated with abnormal heavy metal concentrations (Désirée et al., 2021a, 2021b; Nana et al., 2022).

Proximate agriculture, typical of the Mbam Basin, involves inputs such as phosphate fertilizers containing Ni, Zn, and Cu, as well as pesticides, which can contribute to localized enrichment. The use of domestic residues or aging water supply pipes for irrigation can also introduce lead or copper into the environment (Adewumi et al., 2024; Tudi et al., 2023). The geochemical characteristics of the Mbam, particularly the moderate multi-metal enrichments with an upstream–downstream gradient (Table 2), point to a combination of geological remobilization and anthropogenic inputs (Biami Nya et al., 2025).

In contrast, the Noun River shows distinct profiles, dominated by extremely high concentrations of uranium and vanadium that are uniform along the watercourse. These elements appear less related to human activity and more to a specific geogenic origin, stemming from mineral formations rich in U and V or the weathering of particular volcanic rocks. This interpretation is supported by statistical approaches (PCA, correlations, dendrograms), which clearly separate sites contaminated with U and V (notably in the Noun) from other sites (Figs. 5, 6, 7).

Furthermore, local geochemical conditions such as pH, organic matter content, and sediment grain size (< 75-µm fine fraction) strongly influence the mobility and retention of metals in sediments (Abara A. Biabak et al., 2025; Biami Nya et al., 2025). For example, areas where sediments are finer and clay content is high tended to concentrate Cr, Ni, Zn, and Cu through adsorption. These processes have likely amplified the enrichments observed at downstream Mbam sites, where sediment accumulation is greater. Such mechanisms are comparable to trends reported in mountainous agricultural areas of Morocco and sub-Saharan Africa, where slope farming combined with erosion promotes the downstream accumulation of toxic elements (Sanad et al., 2025; Tindwa & Singh, 2023).

The combined analysis of geochemical data, statistical results, and local socio-economic context suggests that heavy metals in Mbam sediments result from both natural sources and anthropogenic inputs linked to agricultural and artisanal activities. In the Noun River, the predominance of U and V reflects a geogenic signature derived from the weathering of uraniferous granitic and metasedimentary rocks of the Pan-African basement (Toteu et al., 2004). These formations, rich in U–V-bearing minerals such as uraninite, coffinite, and titanomagnetite, naturally release these elements into the fluvial system through chemical weathering and sediment transport. These findings highlight the importance of clearly distinguishing natural contributions from human-induced pollution to effectively guide environmental monitoring and management strategies.

Comparison with other rivers

A comparison of the heavy metal concentrations measured in the sediments of the Mbam and Noun Rivers with those from other river basins at the regional, African, and global scales reveals notable similarities and differences (Table 5). Overall, the levels of Cr, Ni, Cu, Zn, and Pb observed in this study fall within the same range as those reported in other rivers influenced by anthropogenic inputs or controlled by geogenic sources (Tiabou et al., 2024a; Yiika et al., 2024).

In Cameroon, sediments from the Mgoua River in the west (Ndjama et al., 2021) exhibit very high Cu (162 mg/kg), Zn (419 mg/kg), and Pb (248 mg/kg) concentrations substantially exceeding those observed in the Mbam and Noun (Table 5). These values are associated with an intense urban–industrial context marked by both industrial and municipal discharges. Similarly, in the Lékié River in Central Cameroon (Ekoa Bessa et al., 2024), although Cu, Zn, and Pb concentrations were not reported, Ni (71 mg/kg), U (2.4 mg/kg), and Th (10.4 mg/kg) levels are comparable to those in the Noun (Table 4), suggesting a naturally enriched geochemical background in this rural region. Conversely, in the gold-mining zone of Abiete–Toko in Southern Cameroon (Mandeng et al., 2019), Ni levels reach up to 217 mg/kg, with measurable uranium (0.65 mg/kg) (Table 5), reflecting the combined influence of local auriferous formations and artisanal mining. These findings confirm that both geogenic and mining-related inputs contribute significantly to Ni and U enrichment in several Cameroonian basins.

At the African scale, studies in South Africa's Molopo River (Mohajane & Manjoro, 2022) show moderate concentrations of Cr (30–90 mg/kg), Ni (20–60 mg/kg), Cu (20–65 mg/kg), Zn (80–240 mg/

Table 5 Comparative concentrations of trace metals in sediments from some rivers worldwide

Site/basin	Country	Cr	Ni	Cu	Zn	Pb	Cd	U	Th	Context
Noun-Mbam (this study)	Cameroon	6.61–106.66	179.89–128.26	4.74–37.18	26.98–69.09	5.43–29.69	28.99–27.47	13.78–4.34	20.0–20.33	Rural, agriculture
Abiete-Toko (Mandeng et al., 2019)	Cameroon	-	35–217	24	47	5.9	0.13	0.65	-	Gold mining area
Lékié (Ekoa Bessa et al., 2024)	Cameroon	-	71	-	-	-	-	2.4	10.4	Rural, agriculture
Mgoua (Ndjama et al., 2021)	Cameroon	152	65	162	419	248	-	12.7	-	Urban industrial
Xihe (Liu et al., 2021)	China	40–85	25–50	20–45	100–240	20–60	0.3–2	-	-	Urban industrial
Ganga (Siddiqui & Pandey, 2019)	India	60–200	30–85	30–90	100–400	20–120	0.2–1.5	-	-	Basin pollution
Buriganga (Islam et al., 2018)	Bangladesh	297	240	280	-	731	7.7	-	-	Extreme urban
Pearl River (Xiao et al., 2013)	China	60–140	20–70	25–90	90–350	30–85	0.2–1.0	-	-	Urban/rural
Molopo (Mohajane & Manjoro, 2022)	South Africa	30–90	20–60	20–65	80–240	25–70	0.1–0.8	-	-	Urban, traffic
Tannery dumpsites (Oruko et al., 2021)	Kenya/South Africa	205–943	0.6–0.69	0.7–1.9	1.2–1.7	-	1.16	-	-	Tanneries
Manwan Reservoir (Wang et al., 2012)	China	35–140	-	20–85	90–300	25–100	0.2–2.5	-	-	Dam sediments, mining

kg), and Pb (25–70 mg/kg), similar to those in the Mbam (Table 5). In tannery-impacted sites in Kenya and South Africa (Oruko et al., 2021), extremely high Cr levels (205–943 mg/kg) have been reported, far exceeding those in this study (Table 5), illustrating the major impact of industry-specific activities such as leather tanning on sediment Cr enrichment.

In Asia, several urban and industrial rivers also show comparable or higher levels than those in the Cameroonian basins (Table 5). In China's Xihe River Basin (Liu et al., 2021), Cr (40–85 mg/kg), Cu (20–45 mg/kg), and Zn (100–240 mg/kg) levels are close to those in the Mbam. However, Cd (0.3–2 mg/kg) and Hg (0.1–0.5 mg/kg) concentrations are slightly higher. In the Pearl River Basin (Xiao et al., 2013), Zn (90–350 mg/kg), Pb (30–85 mg/kg), and Cu (25–90 mg/kg) levels are similar to those in the Mgoua (West Cameroon; Ndjama et al., 2021) and exceed those in the Mbam and Noun. Likewise, in the Manwan Reservoir (Wang et al., 2012), used for hydropower and affected by mining activities, Zn (90–300 mg/kg) and Pb (25–100 mg/kg) levels are at the higher end of the range observed in this study, while As (10–40 mg/kg) and Cd (0.2–2.5 mg/kg) point to specific enrichments linked to the mining context. Rivers in the Indian subcontinent also exhibit concerning levels. In the Ganges Basin (Siddiqui & Pandey, 2019), Cr (60–200 mg/kg), Cu (30–90 mg/kg), Zn (100–400 mg/kg), and Pb (20–120 mg/kg) are on the same order of magnitude as those reported for the Mbam, while the Buriganga River in Bangladesh (Islam et al., 2018) shows extreme Pb (731 mg/kg), Cr (297 mg/kg), Ni (240 mg/kg), and Cu (280 mg/kg) concentrations far exceeding those measured in Cameroonian rivers. This severe pollution is mainly of urban and industrial origin.

Overall, these comparisons show that the Mbam and Noun Rivers exhibit heavy metal contamination levels generally lower than those found in heavily industrialized environments or areas with industry-specific pressures such as tanneries or intensive mining. However, certain similarities with naturally enriched basins in uranium and nickel, such as the Lékié or Abiete–Toko in Central and Southern Cameroon, suggest that geogenic inputs play a major role in shaping the chemical signature of these sediments (Mandeng et al., 2019; NdemaMbongué et al., 2023; Ekoa Bessa et al., 2024). The marked differences with severely polluted rivers in Asia or North Africa highlight the influence of urbanization and industrialization on contamination levels. These findings demonstrate that although the studied Cameroonian basins are relatively less impacted by massive anthropogenic sources, they display specific enrichments (notably in U and V) that set them apart from many other fluvial systems worldwide. This underlines the need to monitor these metals in areas where geology favors their natural mobilization.

Environmental and health implications

The heavy metals identified in the sediments of the Mbam and Noun Rivers, particularly uranium (U), vanadium (V), chromium (Cr), nickel (Ni), copper (Cu), zinc (Zn), lead (Pb), and tin (Sn), represent a potential threat for both aquatic ecosystems and human health. The combination of geochemical indices (CF, EF, Igeo, PLI) and ecological risk indices (Er, RI), together with human health risk metrics (HI and TCR), provides a coherent picture of the environmental status of both river systems. Overall, the contamination pattern differs between the two basins. In the Mbam River, multi-metal enrichment (Cr, Ni, Cu, Zn, Pb, Sn) is most pronounced at downstream sites, where contamination factors (CF) and pollution load indices (PLI) exceed 1.3–1.5, indicating moderate but widespread anthropogenic and geogenic inputs. In contrast, sediments from the Noun River show systematically elevated concentrations of U and V, with CF values for uranium reaching > 5 at several sites, and ecological risk factors for U ($Er_{(U)}$) exceeding 200 in some cases, which corresponds to a very high ecological risk. These findings are consistent with Igeo and EF values that highlight the specific enrichment of U and V in the Noun, even where other metals remain comparatively low.

From a toxicological standpoint, several sites, especially in the Noun, approach or exceed established sediment quality guideline thresholds such as the TEL (threshold effect level) and, locally, the PEL (probable effect level), notably for U and V. This implies a non-negligible probability of adverse biological effects at those locations. In parallel, the human health assessment shows that the hazard index (HI) for children ranges from ~0.19 to 0.34 across all sites, and the total carcinogenic risk (TCR) lies

between ~10^{-6} and 10^{-4}, which falls within the generally accepted risk range defined by USEPA (1989, 2011). Although these values indicate no immediate critical threat to human health under typical exposure scenarios, they also reveal spatial hotspots where long-term, low-dose exposure particularly for children may be of concern, especially in the downstream Mbam and in uranium-rich sectors of the Noun. These results indicate that (i) the ecological risk is already significant in some parts of the system, particularly due to uranium in the Noun, and (ii) the human health risk is currently within acceptable limits but cannot be considered negligible, especially for vulnerable populations chronically in contact with sediments.

Impact on benthic fauna and aquatic biodiversity

Benthic organisms are directly exposed to sediment-bound metals and are therefore among the first biological groups to show measurable effects. The elevated Er values for U, Pb, Sn, and, locally, Cr and Ni in the Mbam, and the extreme $Er_{(U)}$ values in the Noun (up to > 200), suggest a high to very high ecological pressure on bottom-dwelling invertebrates in several stations. Benthic invertebrates and detritivorous macrofauna are known to bioaccumulate these elements, which can impair physiological function, suppress reproduction, and increase mortality (Chapman et al., 2013; Burton et al., 2017). Cr, Ni, and Pb are particularly toxic to aquatic invertebrates and can induce sublethal stress responses such as oxidative damage and impaired molting (Di Toro et al., 2005; Gao et al., 2021).

In the Noun River, where U and V are systematically enriched, metal-induced histopathological changes in gill and liver tissues of fish and macroinvertebrates have been widely reported in similar uranium-bearing systems (Muscatello et al., 2008; Sarkar et al., 2022). Such chronic stress can alter species composition, reduce overall biodiversity, and destabilize trophic networks. Because these organisms underpin nutrient cycling and sediment reworking, their impairment can trigger cascading effects on higher trophic levels, including fish. The ecological signal is already strong: even where human health indices (HI, TCR) remain within acceptable ranges, the benthic compartment shows signs of elevated risk, especially in uranium-rich reaches of the Noun.

Water quality and transfer to the water column

River sediments act as both sinks and secondary sources of contaminants. Metals deposited in fine alluvial material can be remobilized into the water column during floods, changes in pH or redox conditions, and intensive hydrological events (Förstner & Wittmann, 2012; Burton et al., 2017). This remobilization pathway is particularly relevant in tropical basins with strong seasonal hydrodynamics, such as the Mbam–Noun system. Elements such as Zn, Cu, and especially U are of concern in this context because of their relatively high mobility and their potential to repartition from particulate to dissolved phases under changing geochemical conditions. This process can lead to local exceedances of drinking or irrigation water guidelines (WHO, 2017). In the Noun River, the persistence of elevated U in sediments (CF up to ~5–7) suggests that mobilization during high-flow episodes could episodically elevate dissolved U and V, contributing not only to chemical toxicity but also to increased natural radioactivity in the water column, as has been described in other uraniferous watersheds (Dosseto et al., 2008; Plater et al., 1992). This mechanism is critical because it provides a direct linkage between sediment contamination (sections "Heavy metal concentrations" and "Contamination levels", ecological stress (section "Ecological risks"), and potential chronic human exposure via water use. It also implies that even if water samples meet guideline values during baseflow conditions, high-flow remobilization events may transiently increase exposure risk for downstream communities.

Impacts on agriculture

Floodplains and riparian gardens represent a major interface between contaminated sediments and human activities in the Mbam and Noun basins. Seasonal overbank flooding and irrigation practices promote the transfer of sediment-bound metals into soils used for subsistence crops. This process not only reduces soil fertility through metal stress on plants but also introduces a pathway for metal accumulation in edible tissues (Ali et al., 2019; Zhao et al., 2015). Metals such as Pb, Cr, Ni, and U are known to bioaccumulate in leafy vegetables and tubers, leading to indirect dietary exposure for humans and livestock (Zhang et al., 2020). In systems with persistent U and V enrichment

similar to what is observed along the Noun, uranium has been shown to accumulate in certain crop species, raising chronic kidney toxicity concerns in humans and grazing animals (Bjørklund et al., 2020; Muscatello et al., 2008). This pathway is particularly relevant for children, who, according to our results, systematically display higher HI values (0.19–0.34) than adults at all sampled sites. Even though these HI values remain below the risk threshold of 1, they indicate that repeated, low-level exposure through soil ingestion and crop consumption could become relevant over longer timescales. Thus, the sediments are not only an ecological concern but also a potential source of progressive, food-chain mediated human exposure, especially in communities that rely on floodplain agriculture.

Health risks for human beings

Human exposure to sediment-associated metals in the Mbam and Noun Rivers can occur through multiple pathways: direct ingestion and dermal contact with sediments (e.g., during bathing, artisanal activities, or agriculture), indirect ingestion via contaminated crops, and, to a lesser extent, consumption of locally caught fish and untreated water (Chen et al., 2016; WHO, 2017). The human health assessment carried out in this study quantified both non-carcinogenic (HI) and carcinogenic (TCR) risks for adults and children at each site.

All hazard index values (HI) remained below the critical threshold of 1 for both children and adults, indicating that non-carcinogenic adverse health effects are not expected under typical chronic exposure scenarios. However, HI_child values were consistently higher than HI_adult values, often by an order of magnitude (e.g., ~0.30–0.34 vs ~0.03–0.04 in the Noun), reflecting the higher relative exposure of children due to lower body mass and higher ingestion rates. This pattern is consistent with the known vulnerability of children to neurotoxic and nephrotoxic metals such as Pb, Cr, Ni, and U (Needleman et al., 2004; Jaishankar et al., 2014; Ali et al., 2019).

Carcinogenic risk indices (TCR) also remained within the generally accepted regulatory range (10^{-6}–10^{-4}; USEPA, 1989, 2011), with TCR_child values in the Mbam reaching ~10^{-4} at the most impacted downstream sites. These values approach the upper limit of the acceptable range and therefore identify specific stations in the lower Mbam as priority areas for health-focused monitoring. In the Noun, TCR values were lower (10^{-6}–10^{-5}), suggesting a comparatively reduced lifetime carcinogenic risk despite strong U and V enrichment. This difference reflects contrasting metal signatures: in the Mbam, Cr (assumed as Cr (VI) for conservative screening), Ni, and Cu dominate exposure; and in the Noun, U is the main driver of both ecological and potential human health concerns due to its chemical (nephrotoxic) rather than radiological toxicity (Bjørklund et al., 2020).

Metals of particular concern include Pb, which is neurotoxic and impairs cognitive development in children even at low doses (Needleman et al., 2004; WHO, 2017); Cr (VI) and Ni, which are recognized carcinogens (Loomis et al., 2014); and U, which can cause renal and skeletal damage under chronic exposure (Bjørklund et al., 2020). Although measured HI and TCR values remain within acceptable ranges, the proximity of some sites, especially in the downstream Mbam to advisory thresholds, suggests that the current situation is best described as "controlled but vulnerable," rather than intrinsically safe.

Long-term ecological and socio-environmental consequences

Heavy metals in river sediments are persistent, poorly degradable, and capable of long-term recycling between solid and dissolved phases (Förstner & Wittmann, 2012). This persistence has several consequences. First, extended exposure of aquatic organisms leads to chronic ecological degradation, including altered community composition, reduced biodiversity, and the loss of functional groups essential for nutrient cycling (Chapman et al., 2016; Dalu et al., 2021). This is particularly relevant in the Noun, where uranium-driven ecological risk ($Er_{(U)}$ and RI) is already high to very high at several sites, indicating ongoing biological stress even in the absence of acute toxicity to humans.

Second, sustained exposure of riparian communities, through water use, agriculture, and fisheries, establishes slow but cumulative human health pressure. Similar river systems in Africa and Asia have

shown that long-term sediment contamination is associated not only with ecological decline but also with elevated incidence of chronic diseases, including neurological, renal, and cardiovascular disorders (Ali et al., 2019; Islam et al., 2015; Jaishankar et al., 2014). The pattern observed in the Mbam–Noun system matches this concern: ecological indicators (Er, RI) already signal stress, and health indicators (HI, TCR), although within acceptable limits, are trending higher at certain sites and for sensitive groups (children).

Third, these processes have socio-economic implications. Declines in sediment quality can impair fisheries, reduce agricultural productivity in floodplains, and increase the long-term cost of providing safe water. In systems where river sediments are directly handled (e.g., artisanal mining, brickmaking, riverbank gardening), the exposure pathway is direct and continuous. This means that even moderate index values today can translate into significant health burdens over decadal scales if no mitigation is implemented.

In summary, metal contamination in the Mbam and Noun Rivers is not only a geochemical issue but also a coupled ecological and public health problem. The ecological risk is already substantial in uranium-enriched reaches of the Noun, and human health risks remain mostly within regulatory acceptability, localized hotspots, particularly in the downstream Mbam warrant priority monitoring, exposure reduction measures, and integration of health surveillance into watershed management strategies (Burton et al., 2017; Ali et al., 2019).

Conclusions

This study provides the first integrated assessment of heavy metal contamination and ecological and health risks in the alluvial sediments of the Mbam and Noun Rivers, Central Cameroon. The main findings are as follows:

1. Spatial variations in metal concentrations reflect the combined influence of natural weathering of uraniferous basement rocks and localized anthropogenic inputs from agriculture, artisanal mining, and small-scale urban discharges.
2. Concentrations of Cr, Ni, Cu, Zn, Pb, and Sn in the Mbam, and U and V in the Noun, exceeded upper continental crust (UCC) values and international benchmarks (TEL, PEL, ERL, ERM). Uranium and vanadium enrichments are distinctive geochemical anomalies linked to local lithology.
3. Ecological indices (Er, RI) reveal moderate contamination in the Mbam and considerable to very high risk in the Noun due to uranium. Human health indices ($HI < 1$, $TCR = 10^{-6}$–10^{-4}) indicate no immediate concern but higher vulnerability for children and downstream communities.
4. Continuous monitoring, control of artisanal mining and agrochemical runoff, and targeted sediment remediation are crucial. Local populations should avoid direct sediment use for agriculture, adopt safer water sources, and participate in community-based river management initiatives to reduce exposure and preserve ecosystem health.

This study establishes a scientific basis for understanding heavy metal dynamics in tropical river systems and supports the development of sustainable watershed management strategies in Central Africa.

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